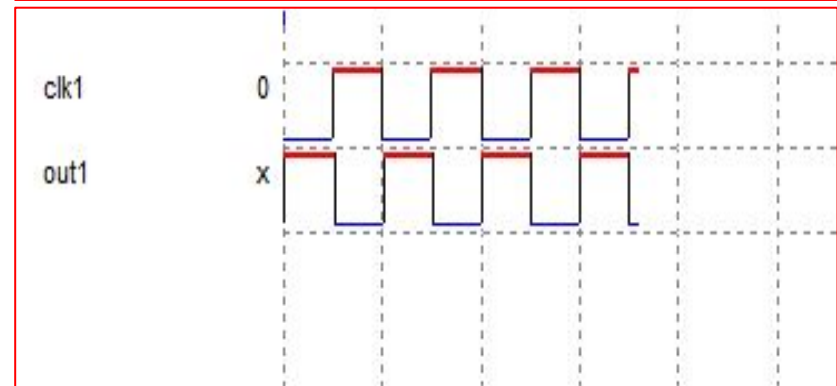
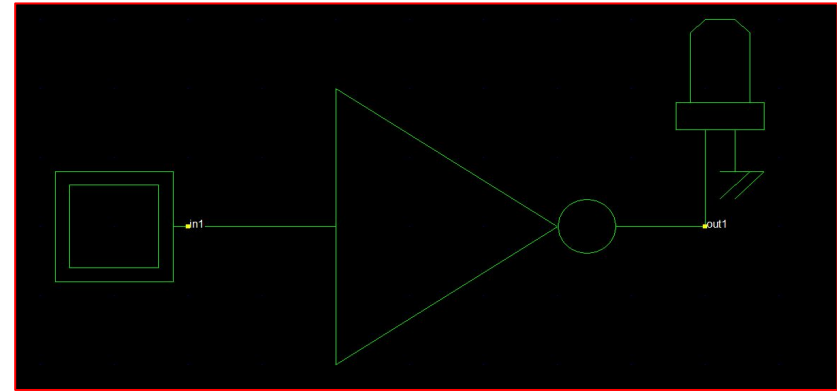
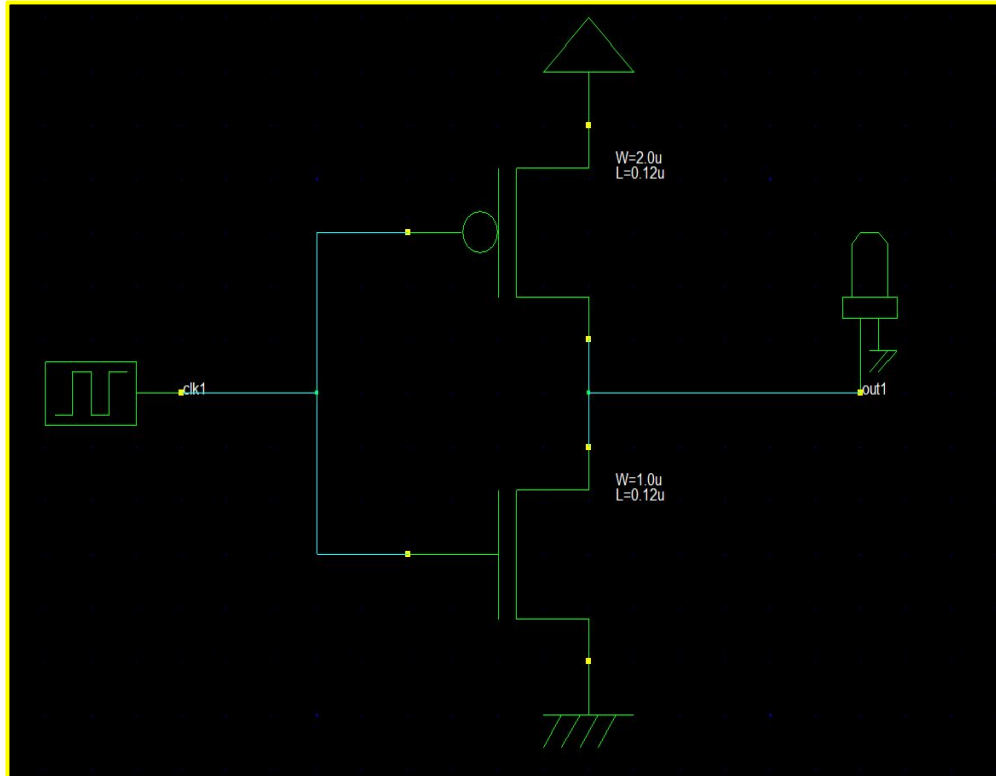


CMOS VLSI And Aspects of ASIC Design LAB

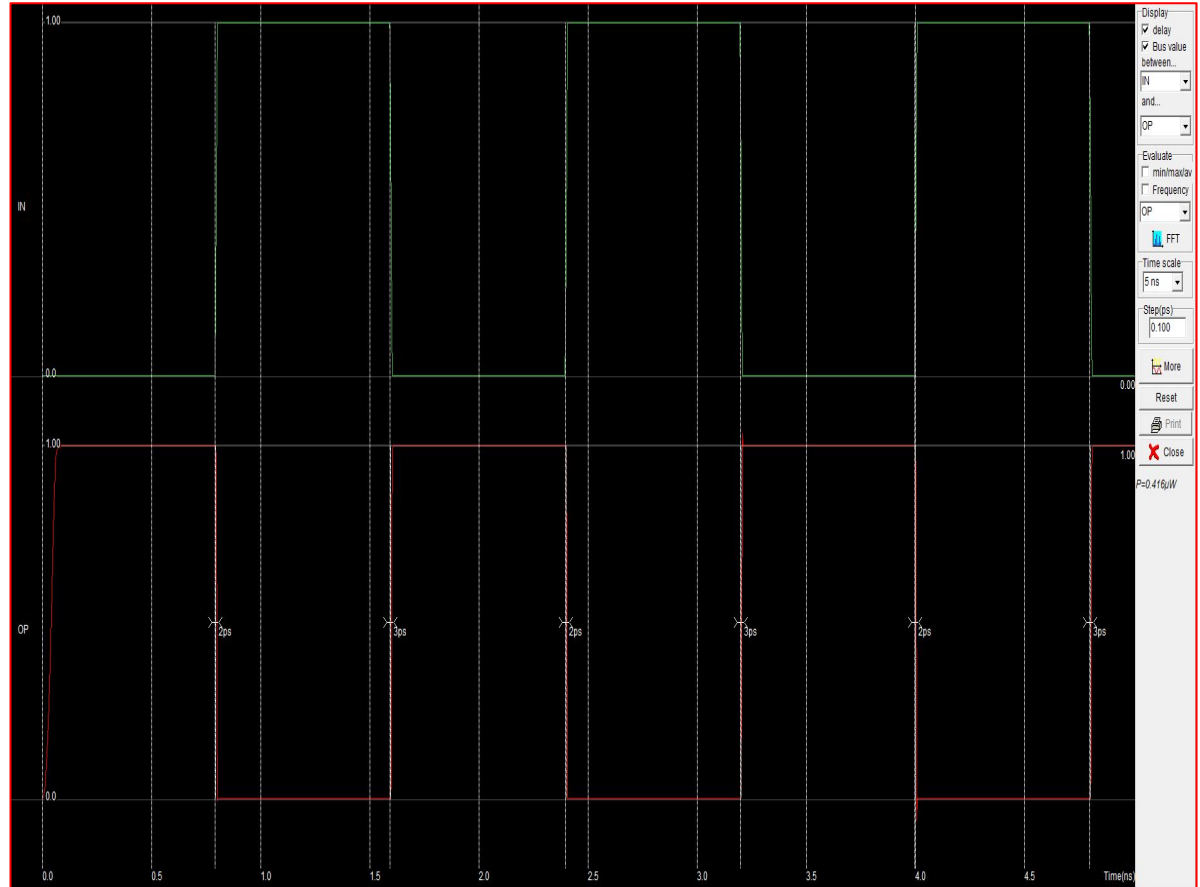
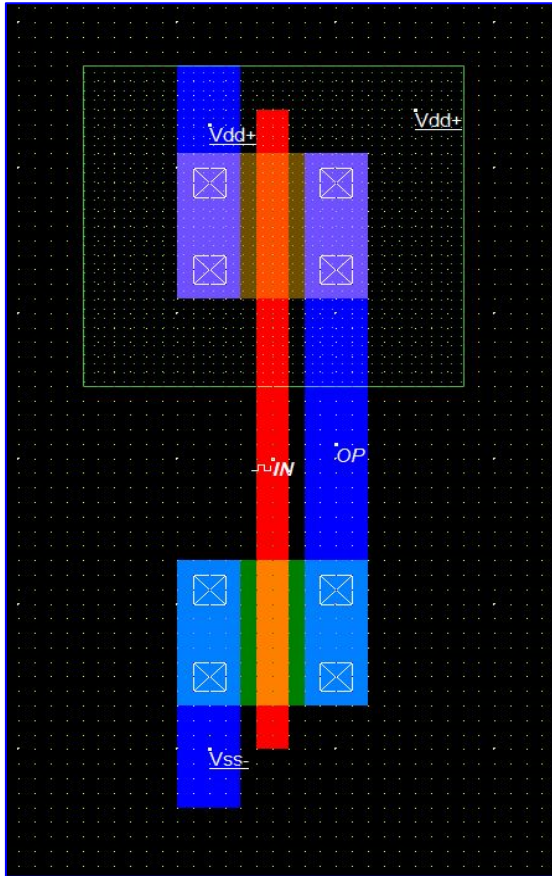
Partha Pratim Mishra (260240133025)
Shraddha Sahoo (260240133032)

CMOS INVERTER

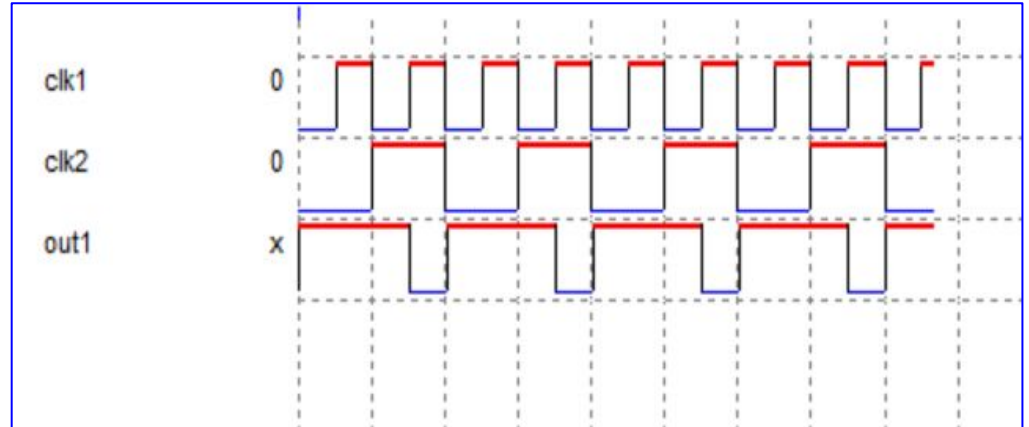
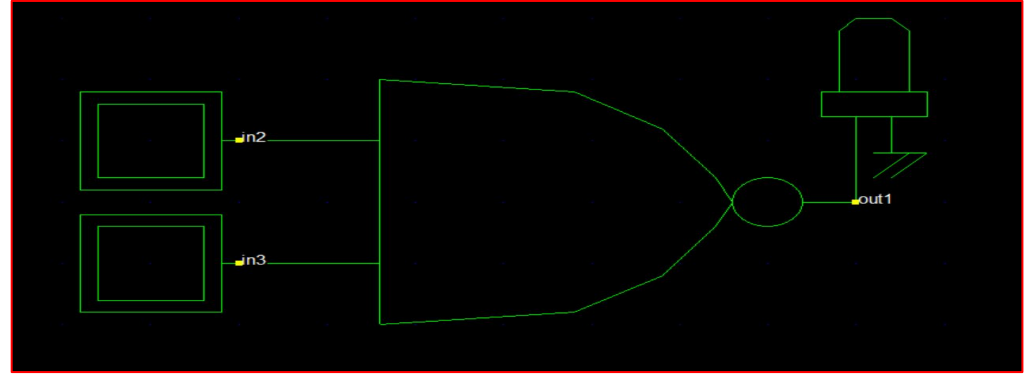
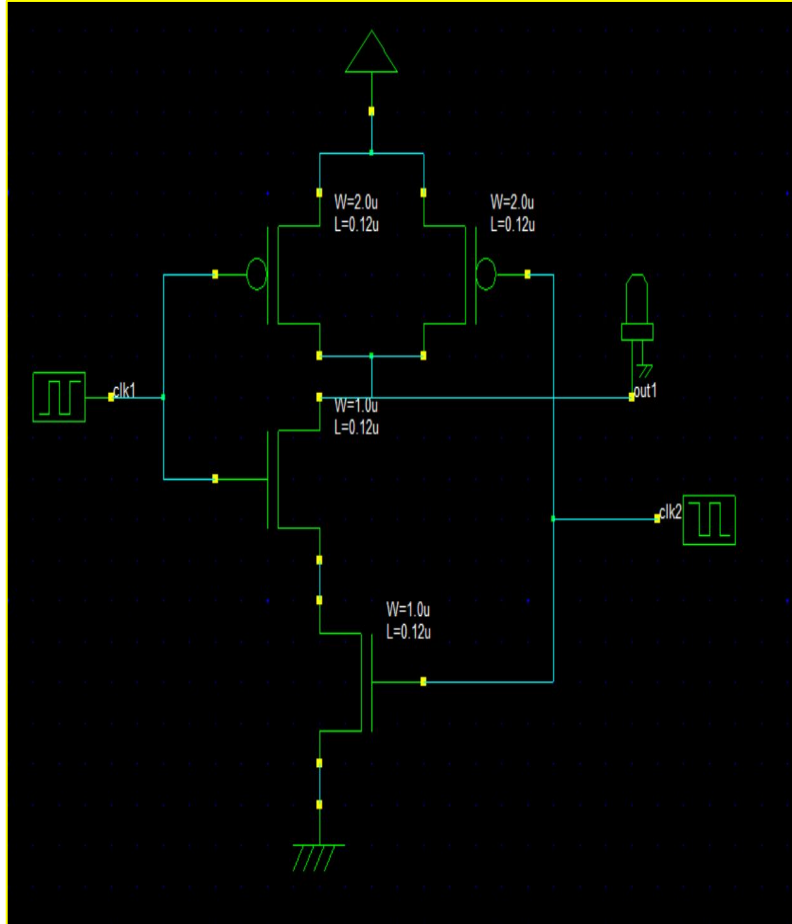


CMOS Inverter Layout

In: timeperiod = 0.090ns + 0.090 ns = 0.18ns

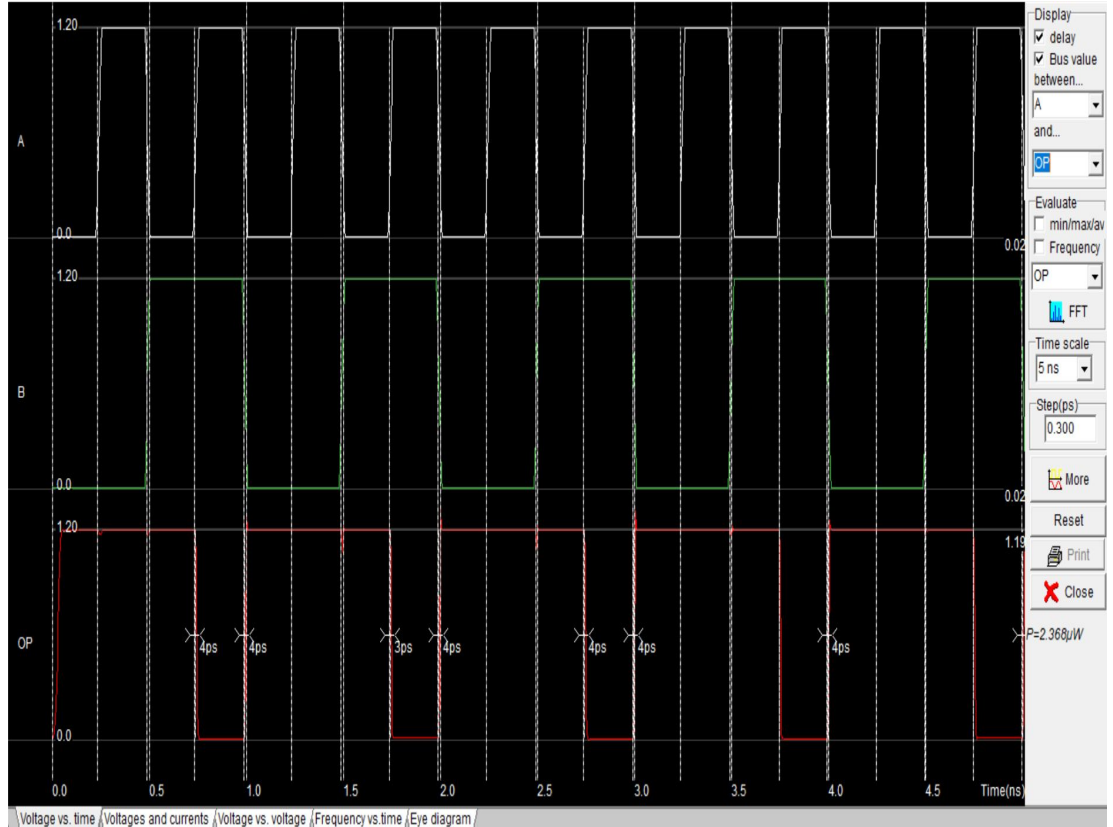
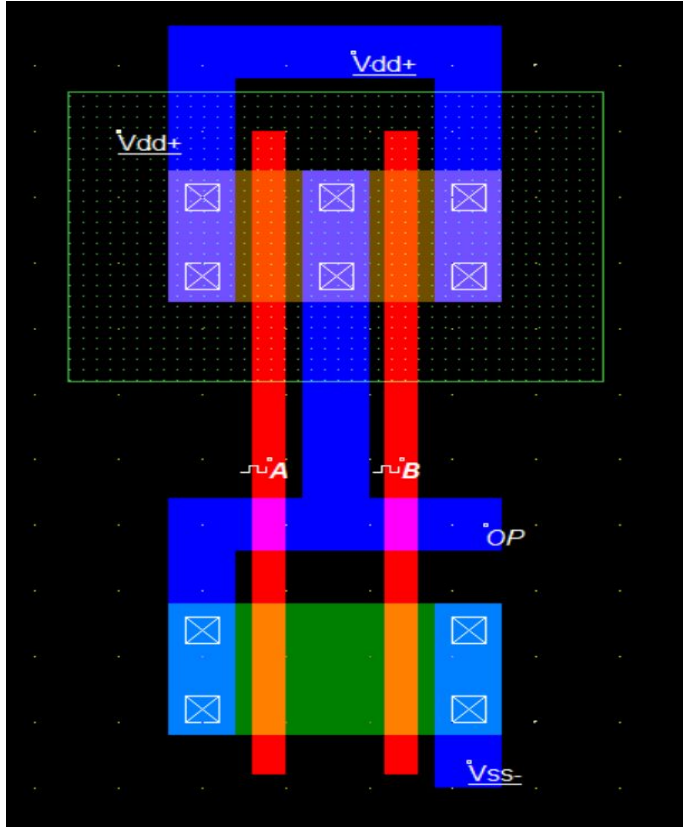


CMOS NAND

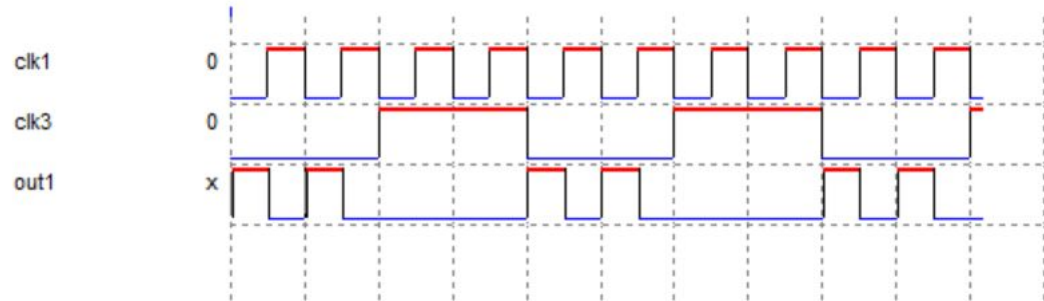
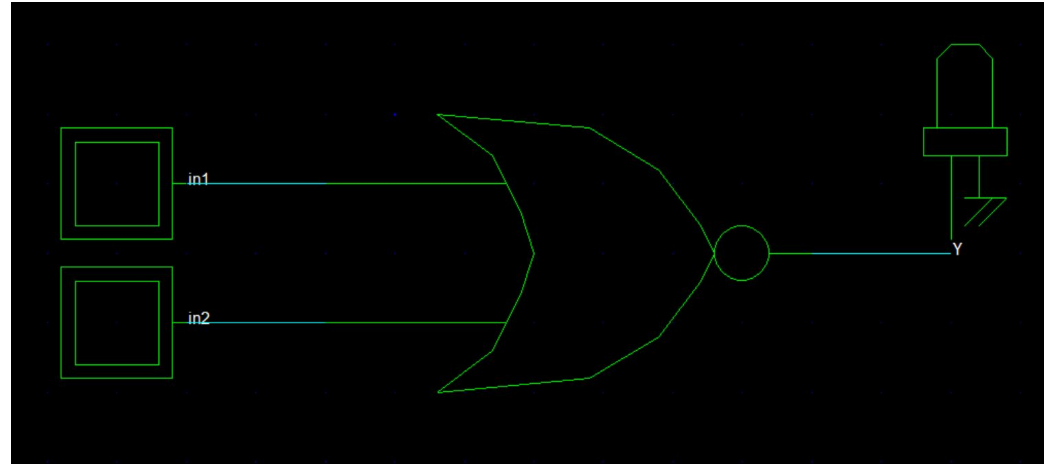
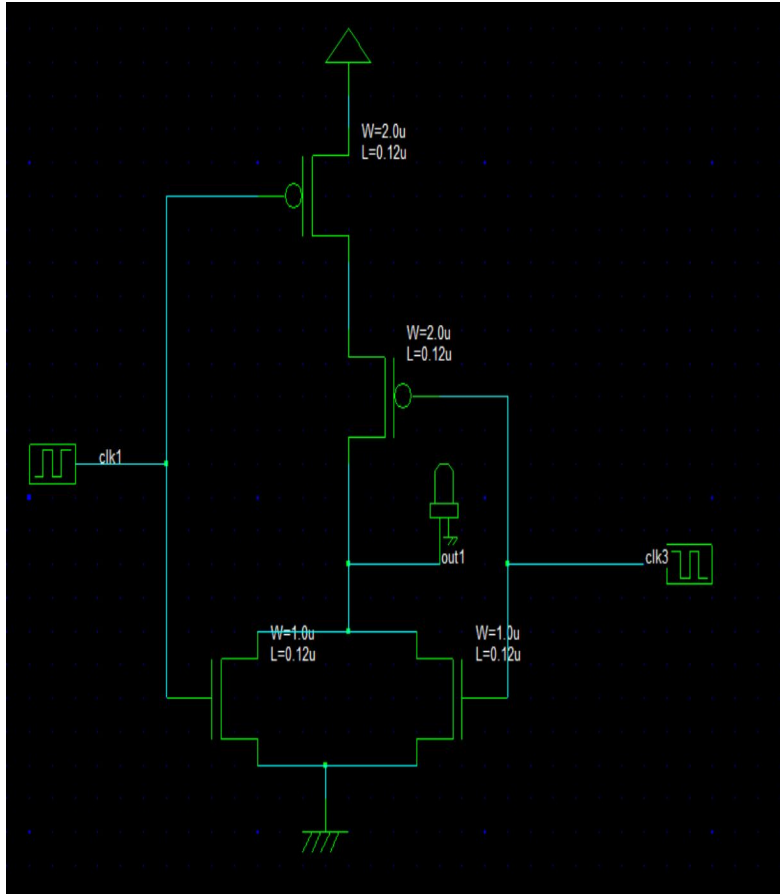


CMOS NAND LAYOUT

A: time period = $0.225\text{ns} + 0.225\text{ns} = 0.450\text{ns}$ B: time period = $0.475\text{ns} + 0.475\text{ns} = 0.950\text{ns}$

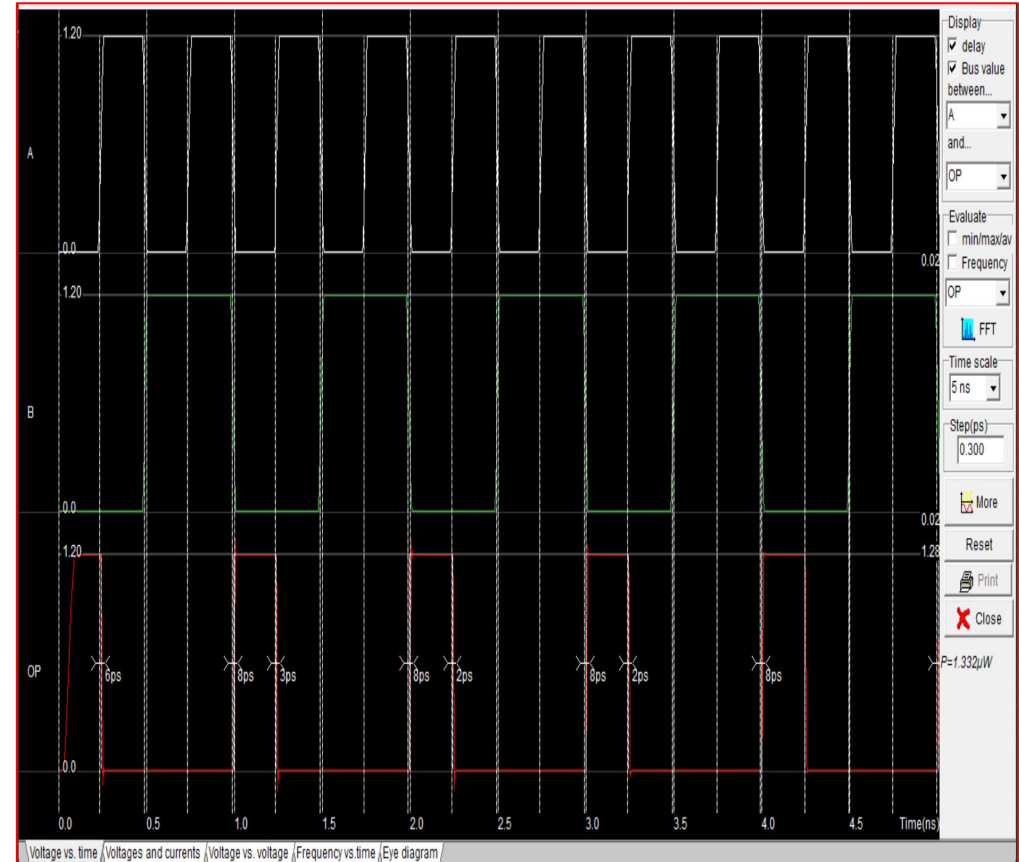
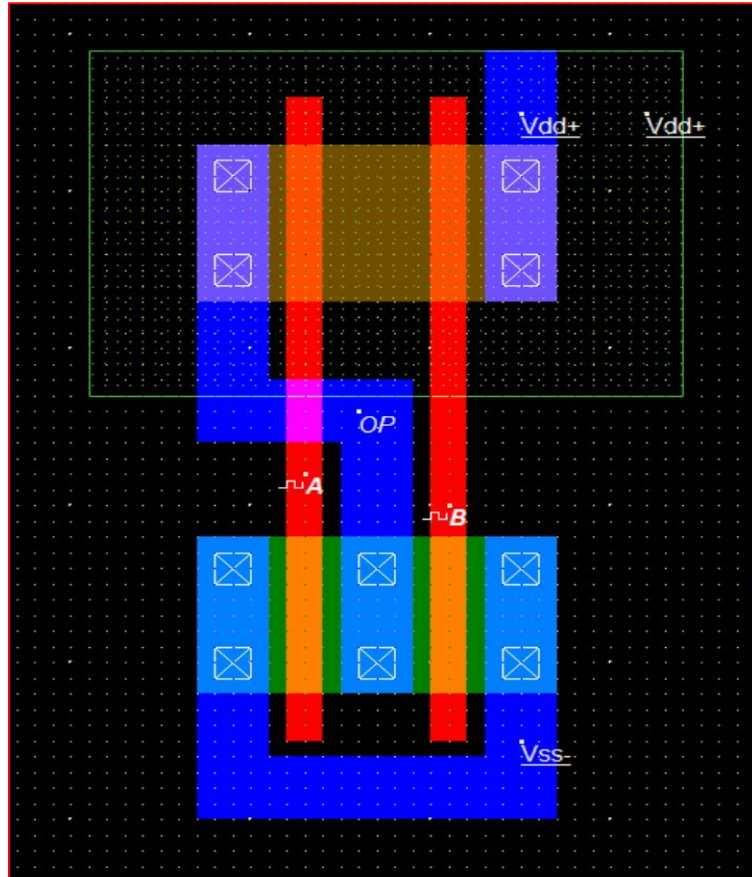


CMOS NOR Schematic

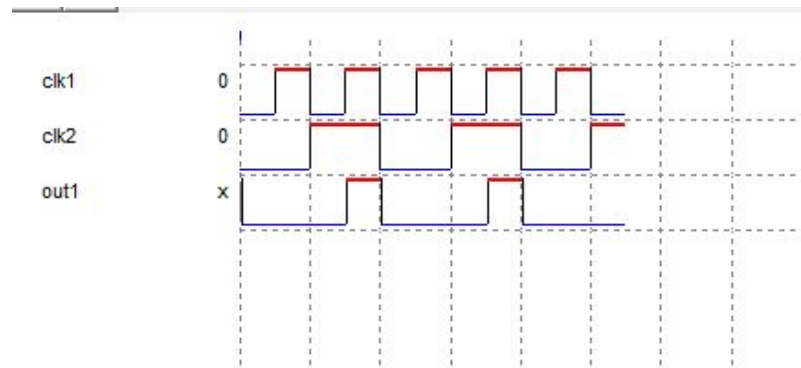
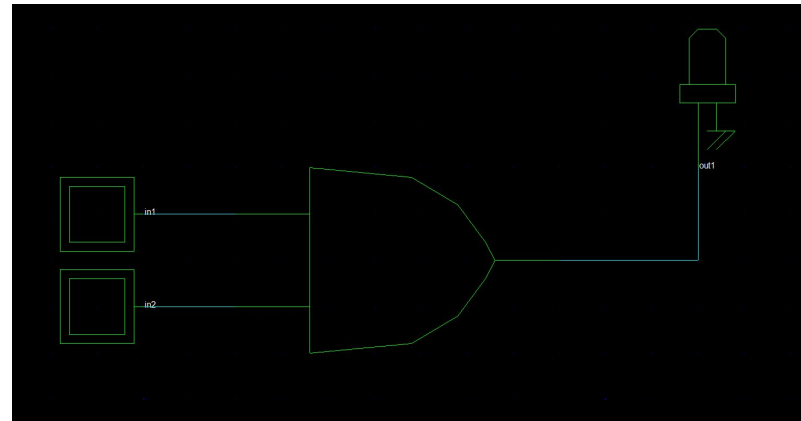
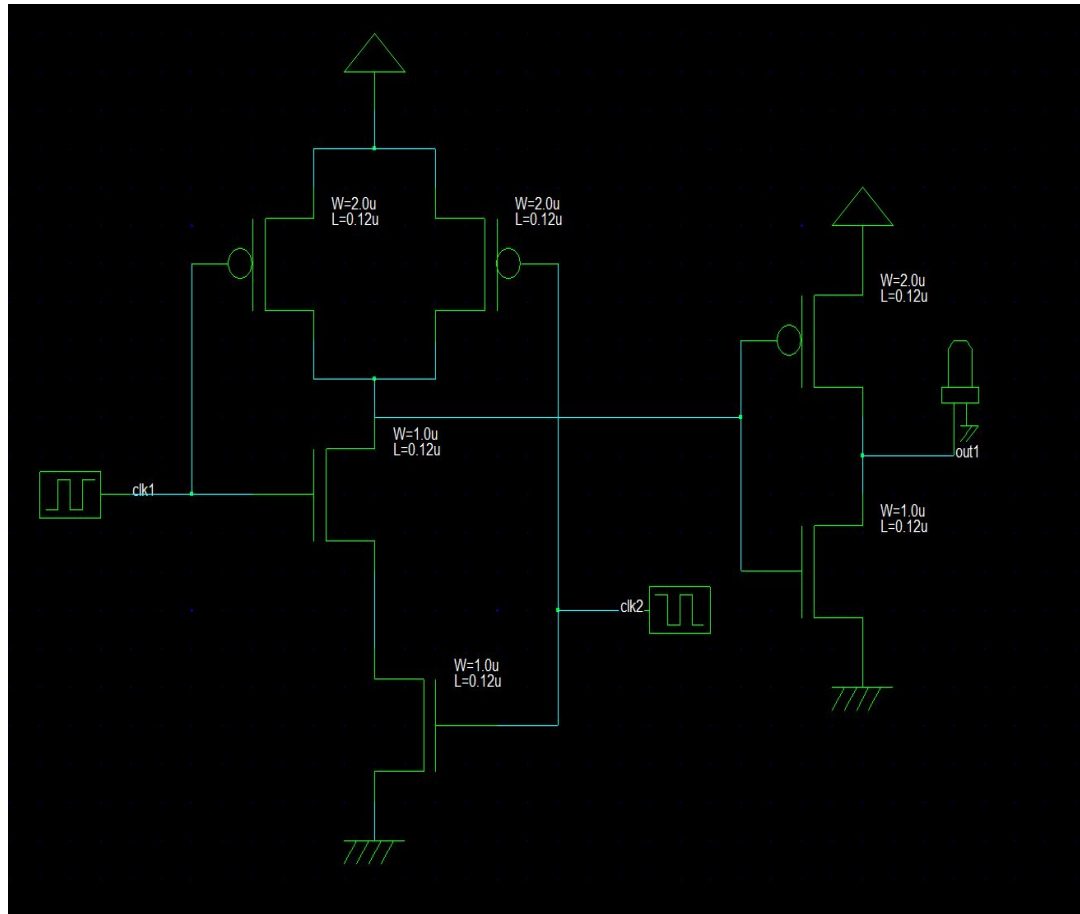


CMOS NOR Layout

A: time period = $0.225\text{ns} + 0.225\text{ns} = 0.450\text{ns}$ B: time period = $0.475\text{ns} + 0.475\text{ns} = 0.950\text{ns}$

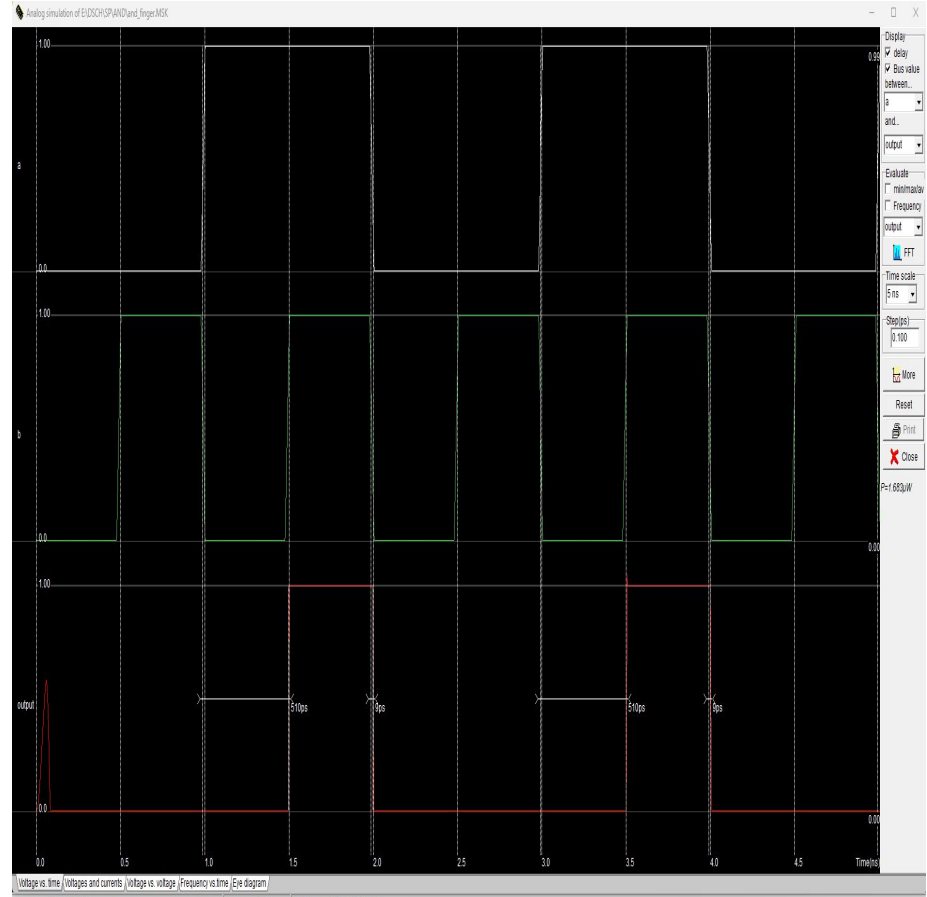
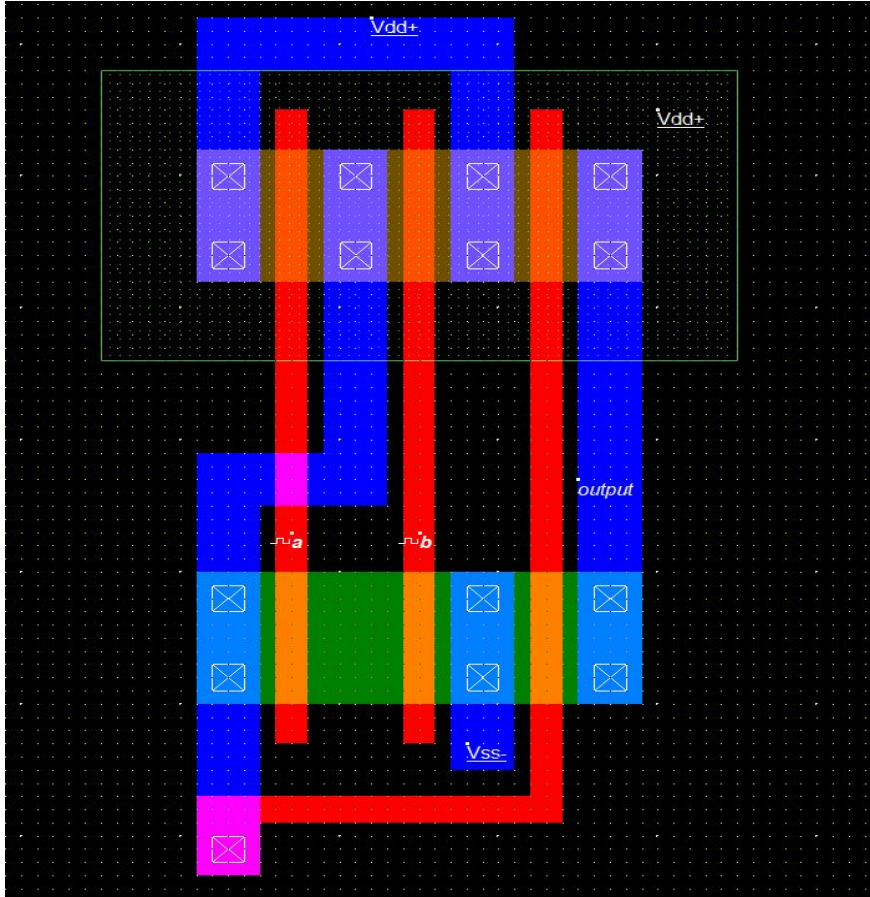


CMOS AND Schematic

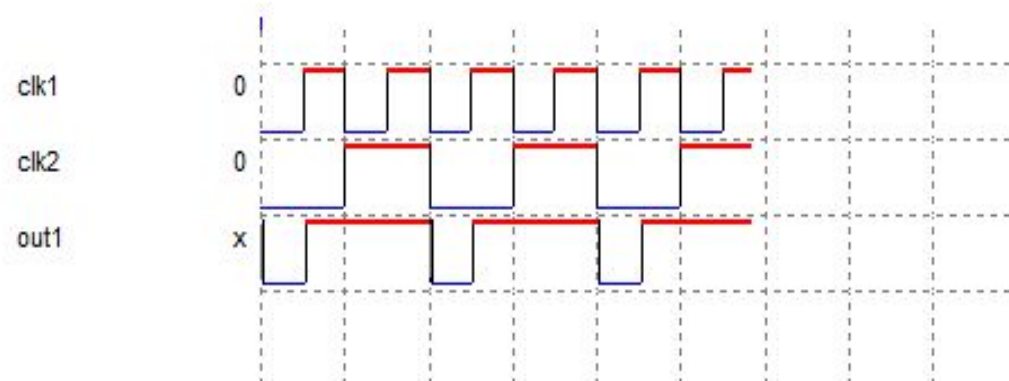
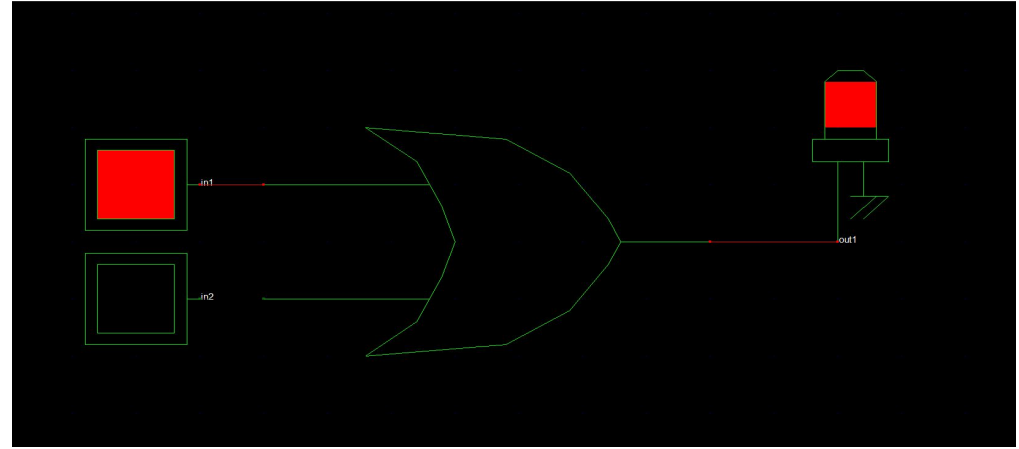
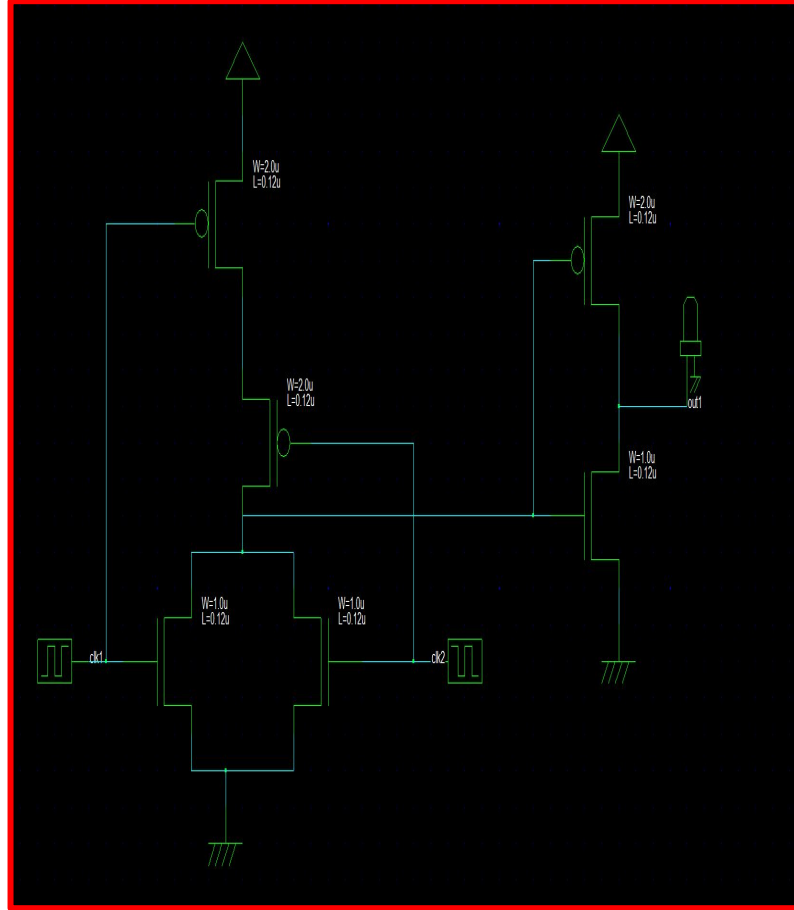


CMOS AND Layout

A: $0.975\text{ns} + 0.975\text{ns} = 1.950\text{ns}$ B: $0.475\text{ns} + 0.475\text{ns} = 0.95\text{ns}$

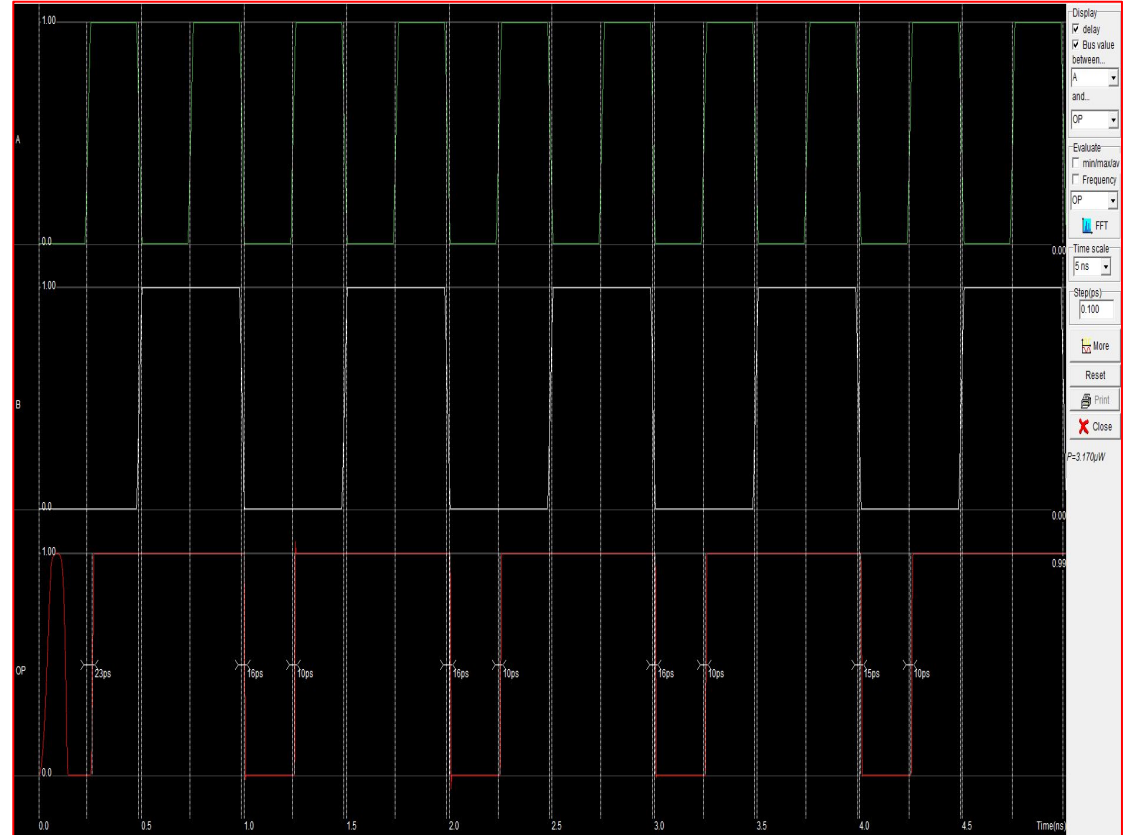
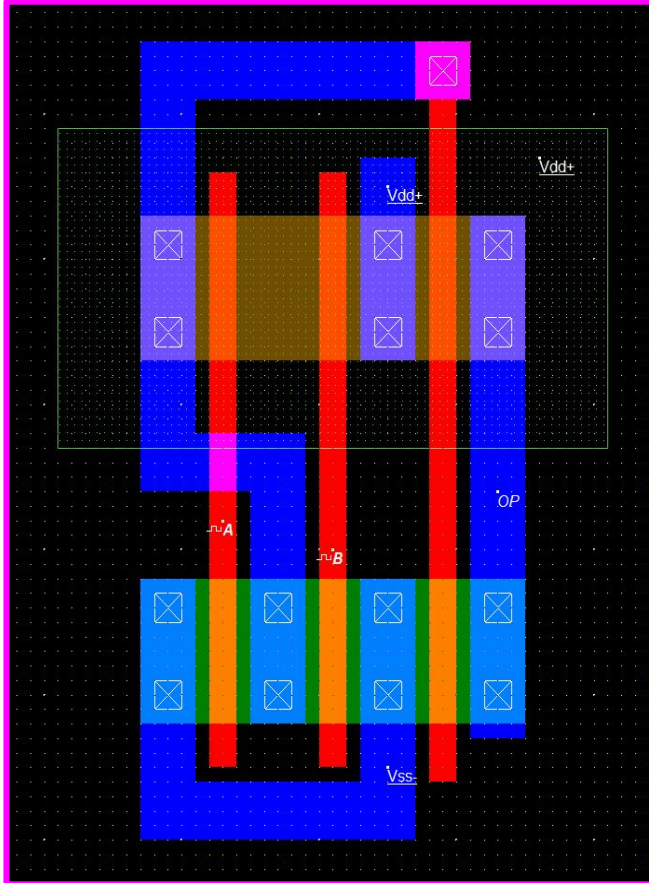


CMOS OR Schematic

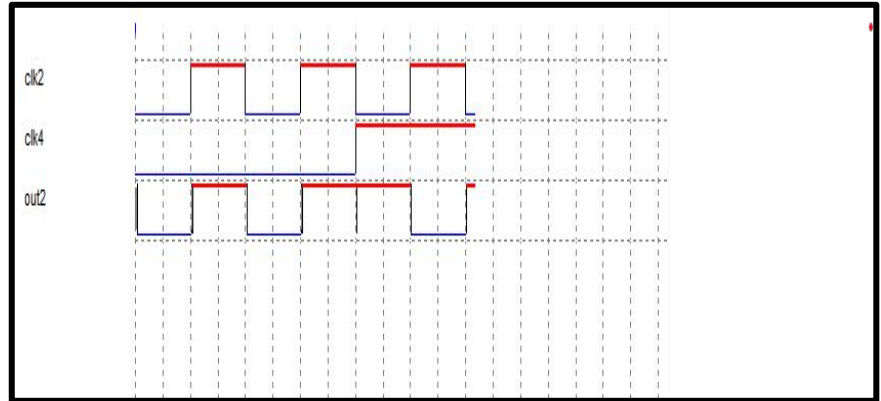
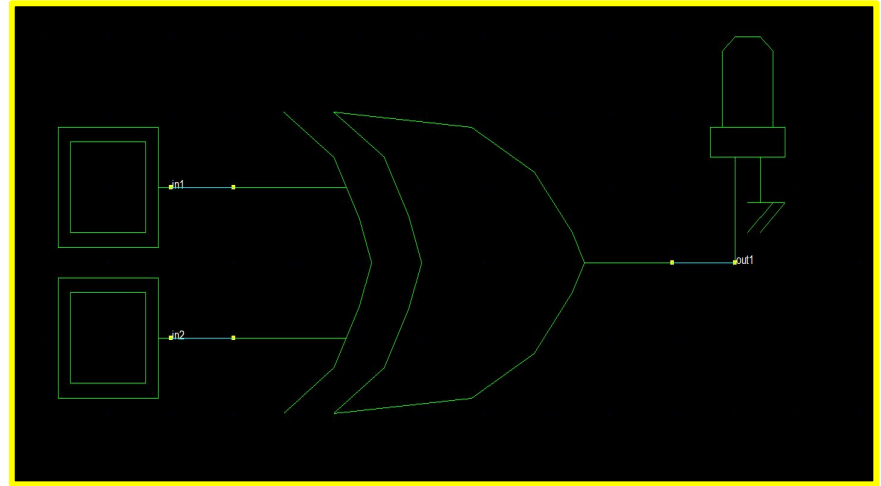
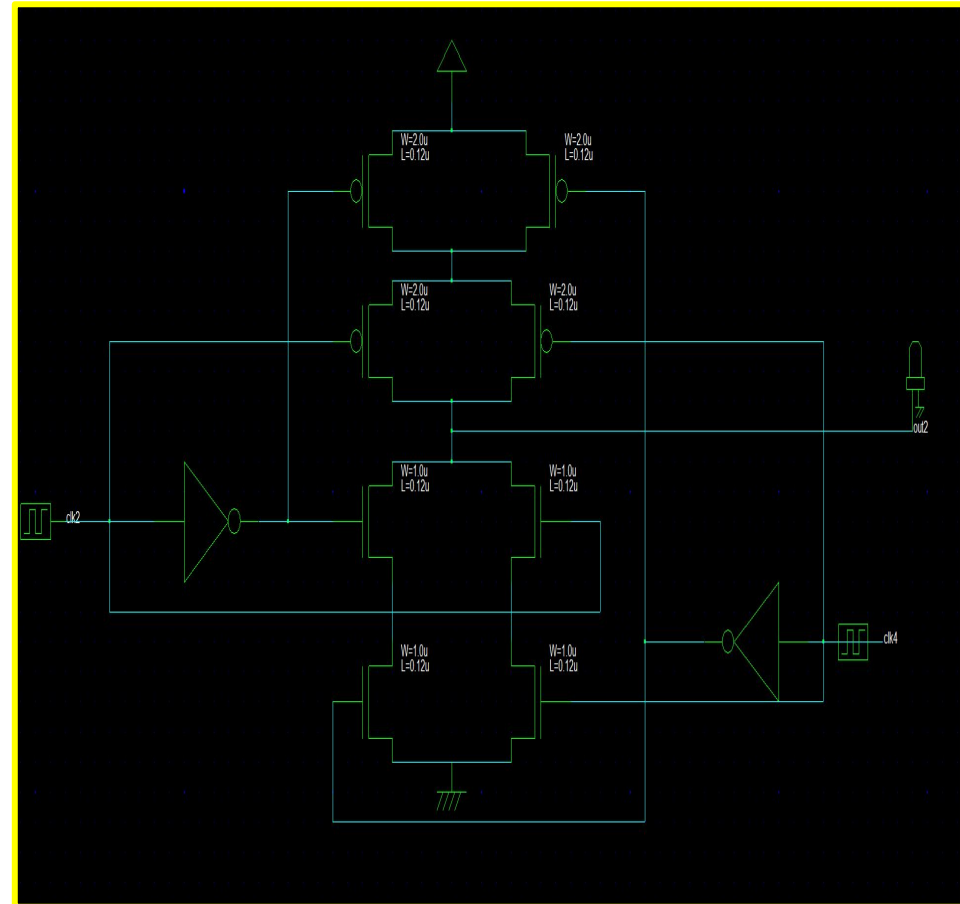


CMOS OR Layout

A: $0.225\text{ns} + 0.225\text{ns} = 0.450\text{ns}$ B: $0.475\text{ns} + 0.475\text{ns} = 0.950\text{ns}$

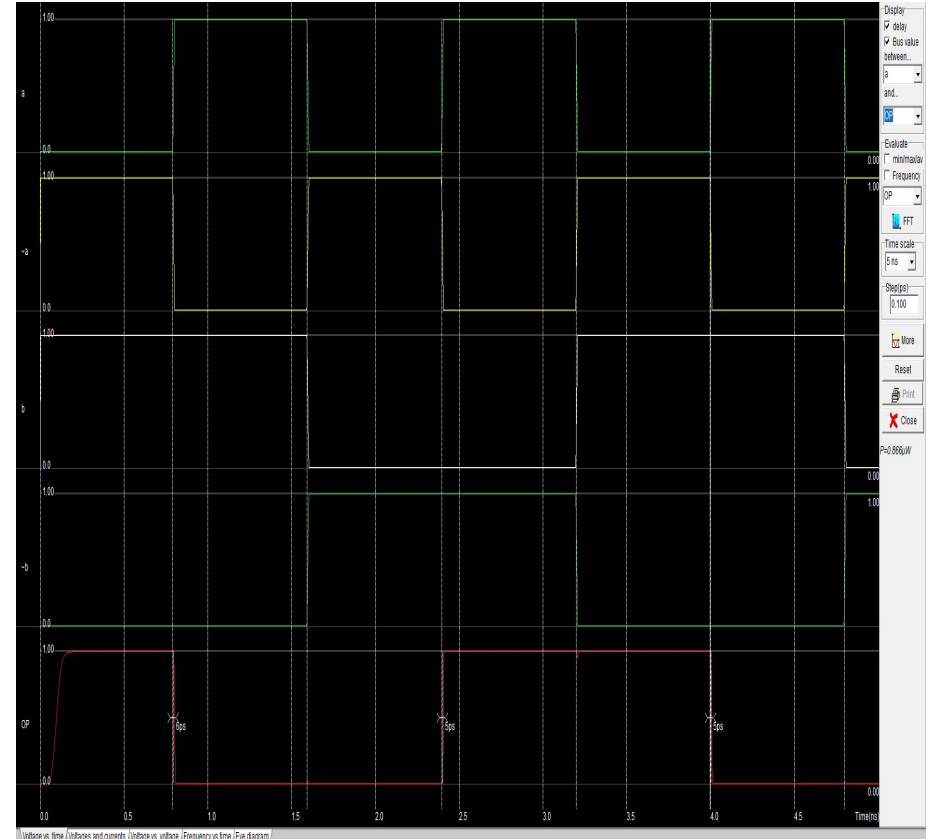
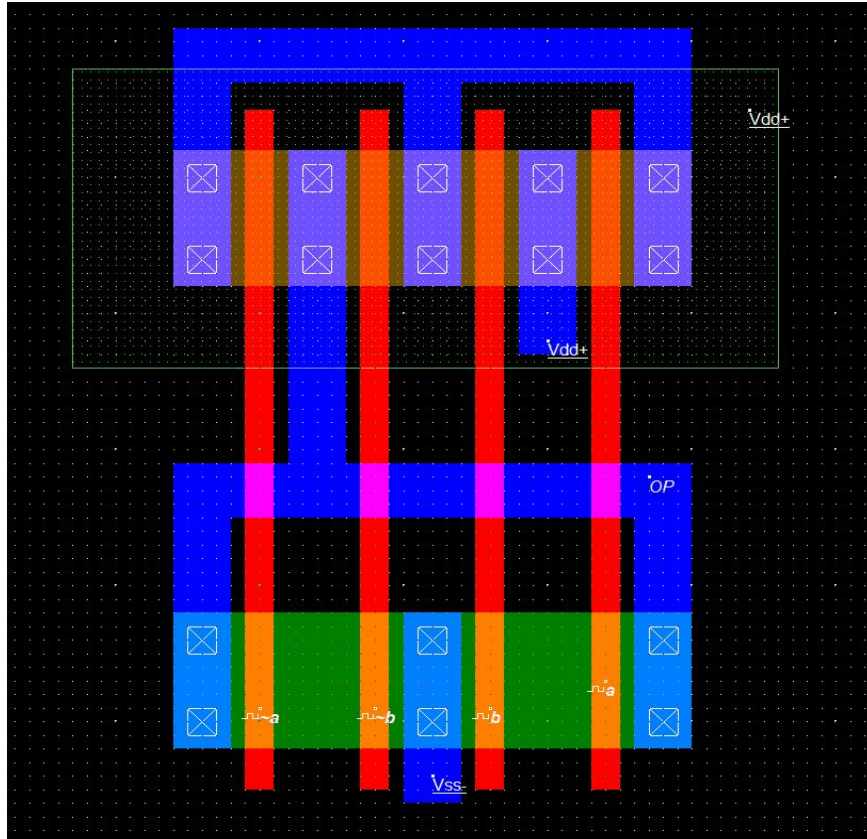


CMOS XOR Schematic

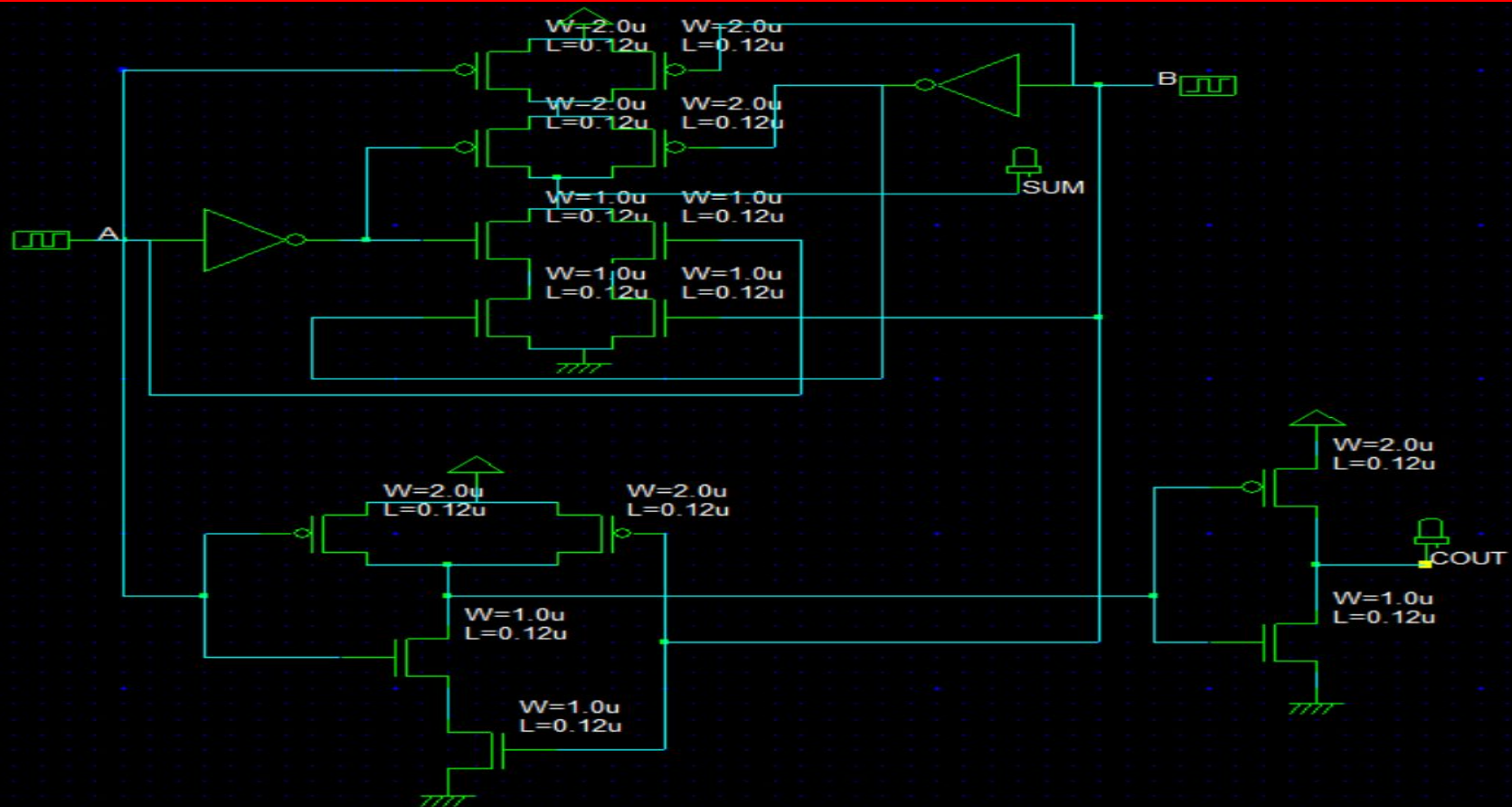


CMOS XOR Layout

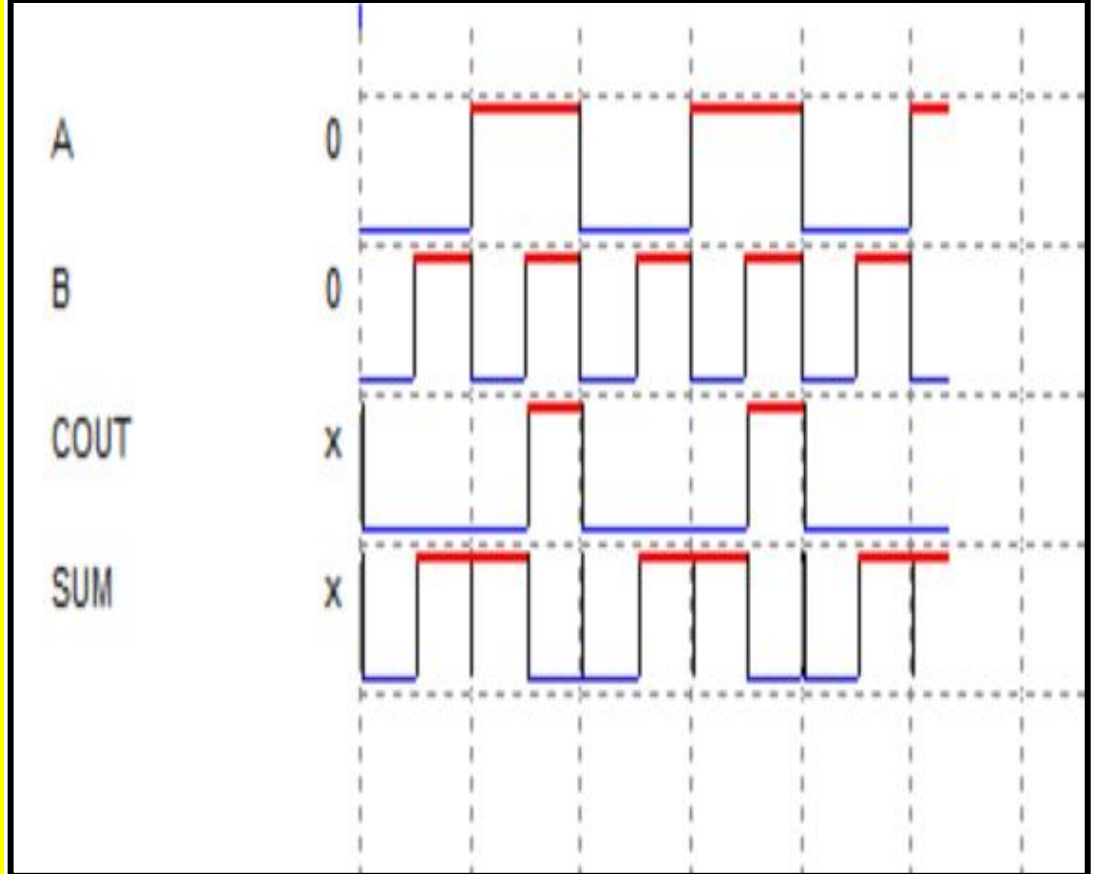
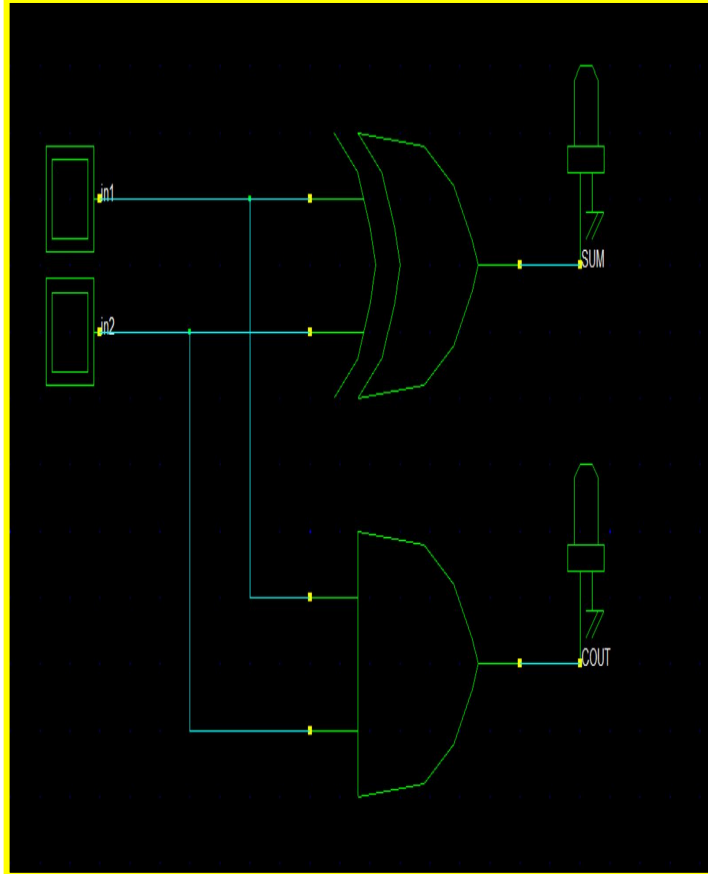
A: $0.790 + 0.790\text{ns} = 1.680\text{ns}$ B: $1.590 + 1.590\text{ns} = 3.180\text{ns}$



CMOS Half Adder Schematic

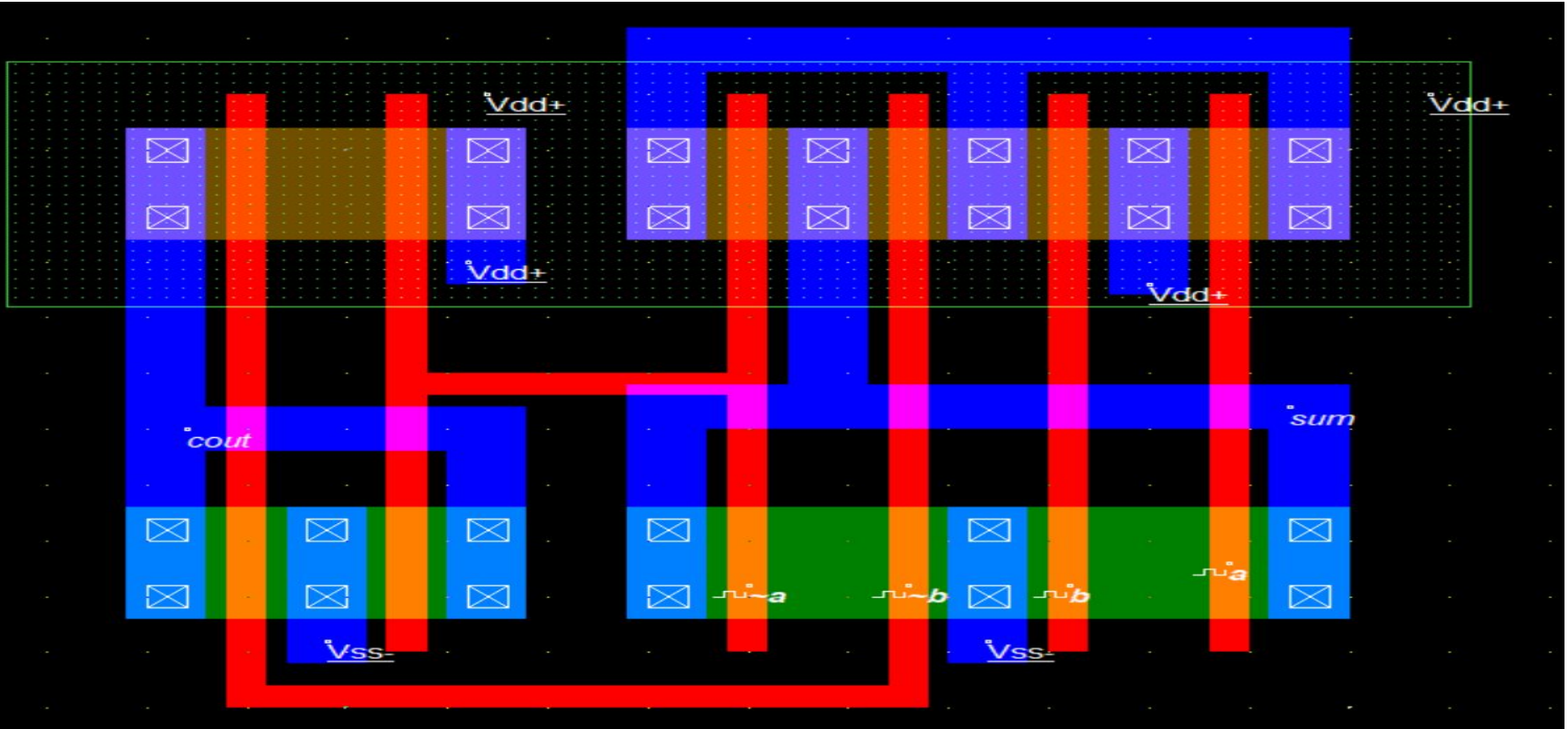


CMOS Half Adder Gate and Transient Analysis

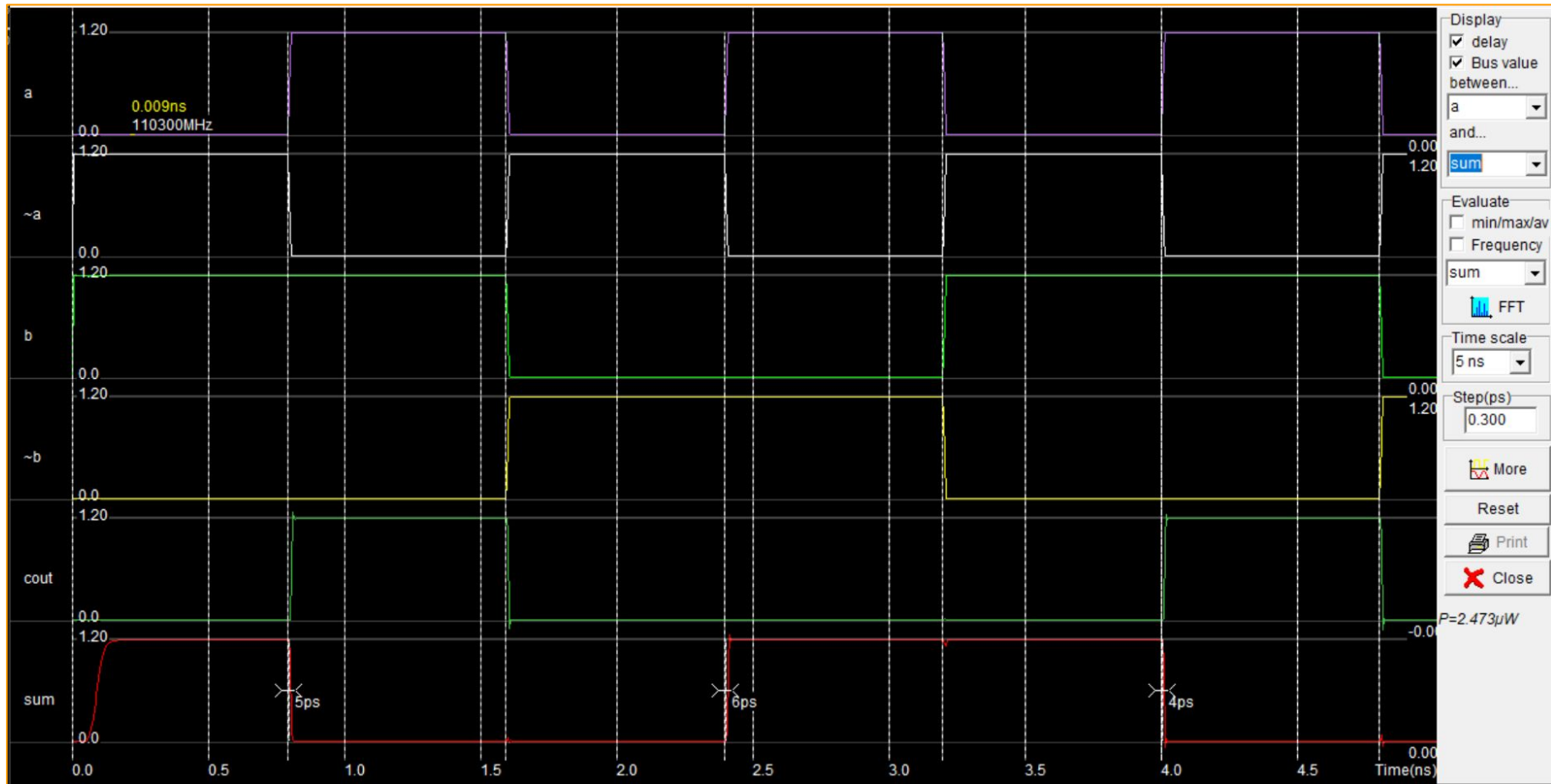


CMOS HA Layout

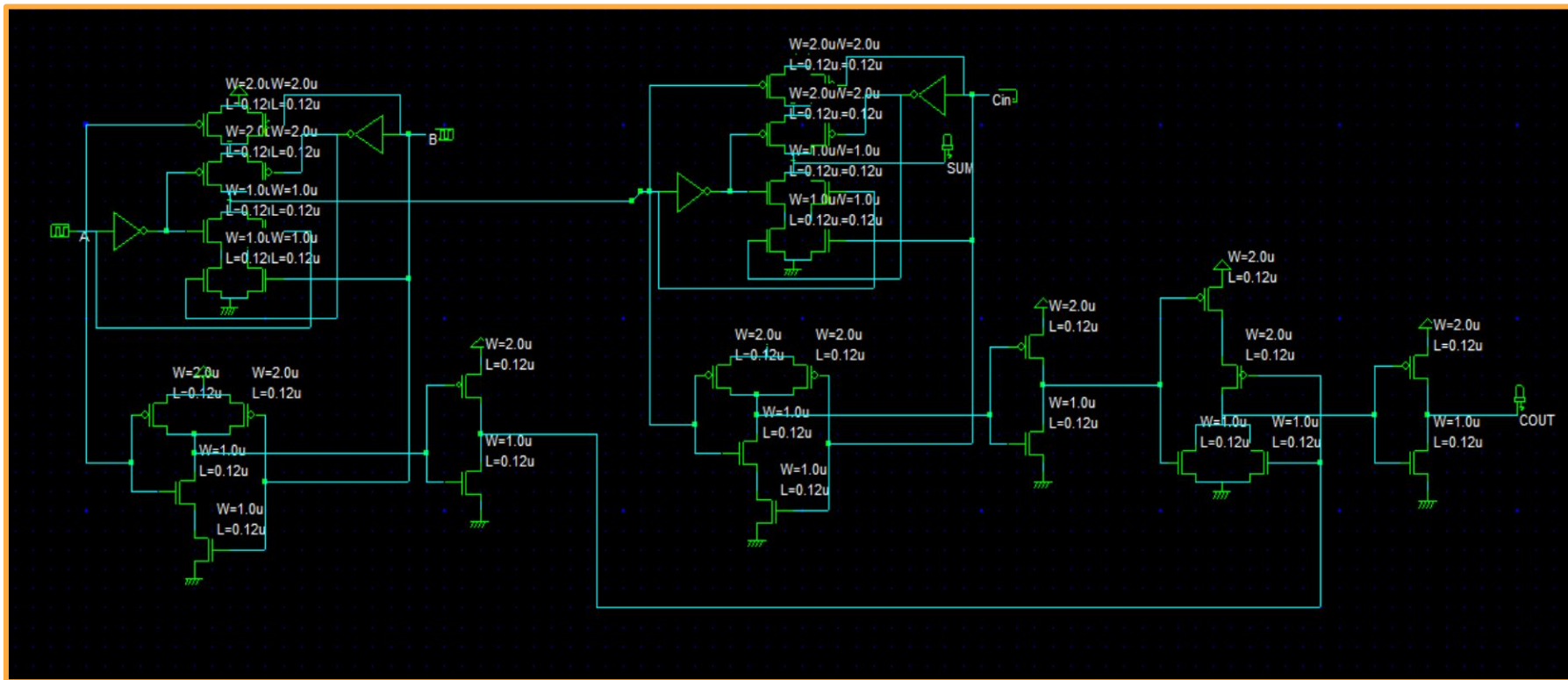
A: $0.790\text{ns} + 0.790\text{ns} = 1.680\text{ns}$ B: $1.590\text{ns} + 1.590\text{ns} = 3.180\text{ns}$



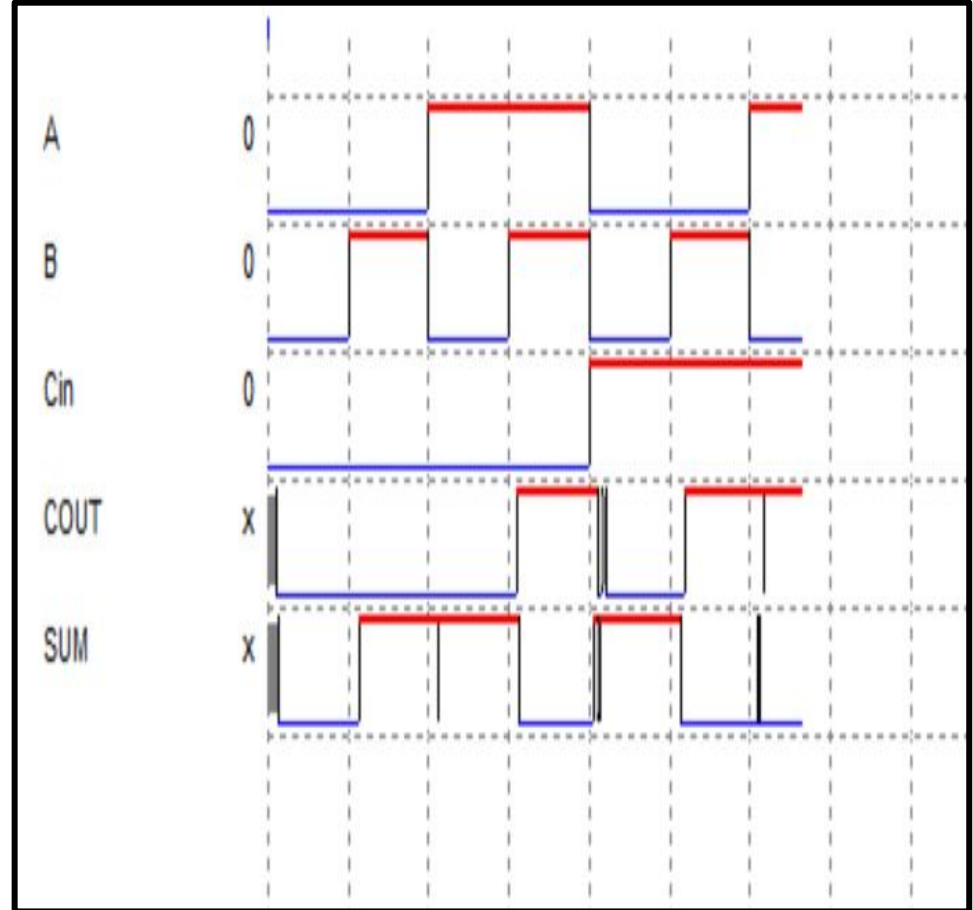
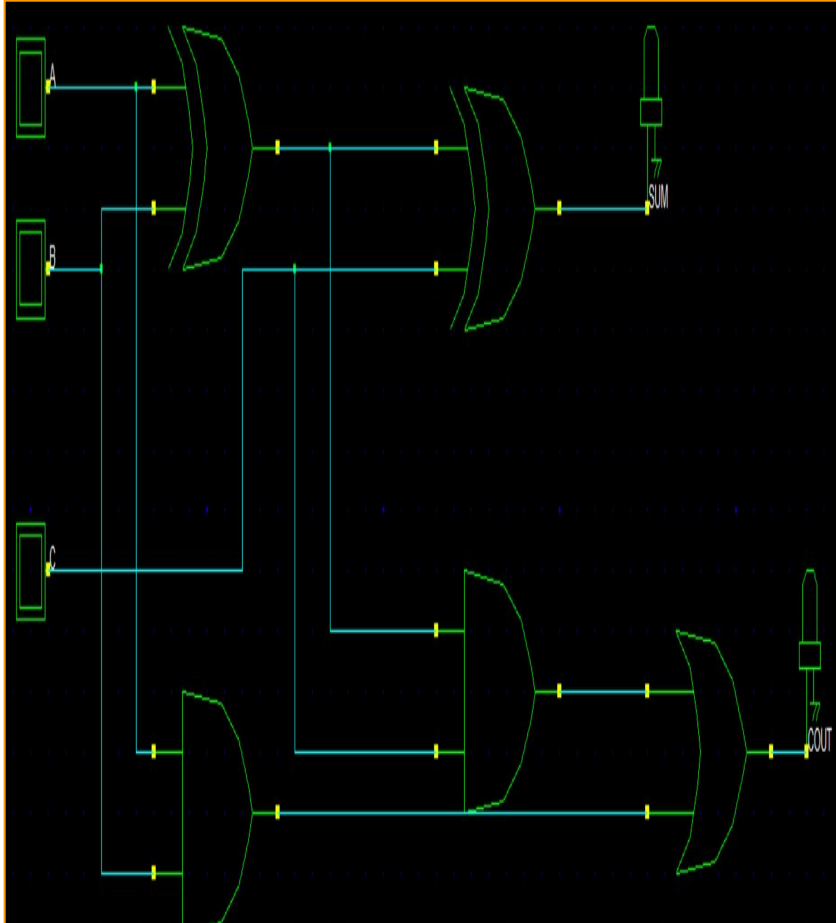
CMOS_HA_Layout Transient Analysis



CMOS Full Adder Schematic

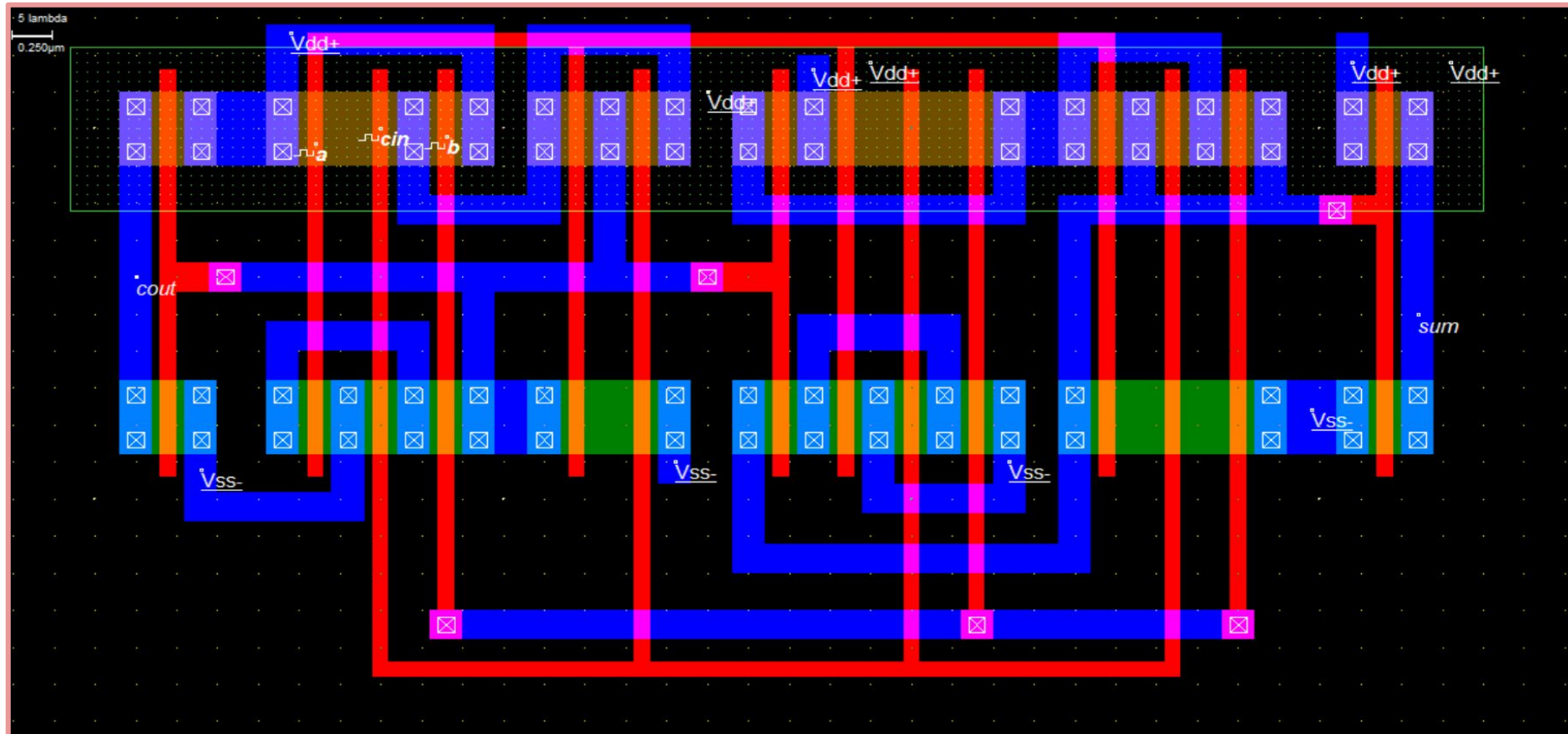


CMOS Full Adder Gate and Transient Analysis

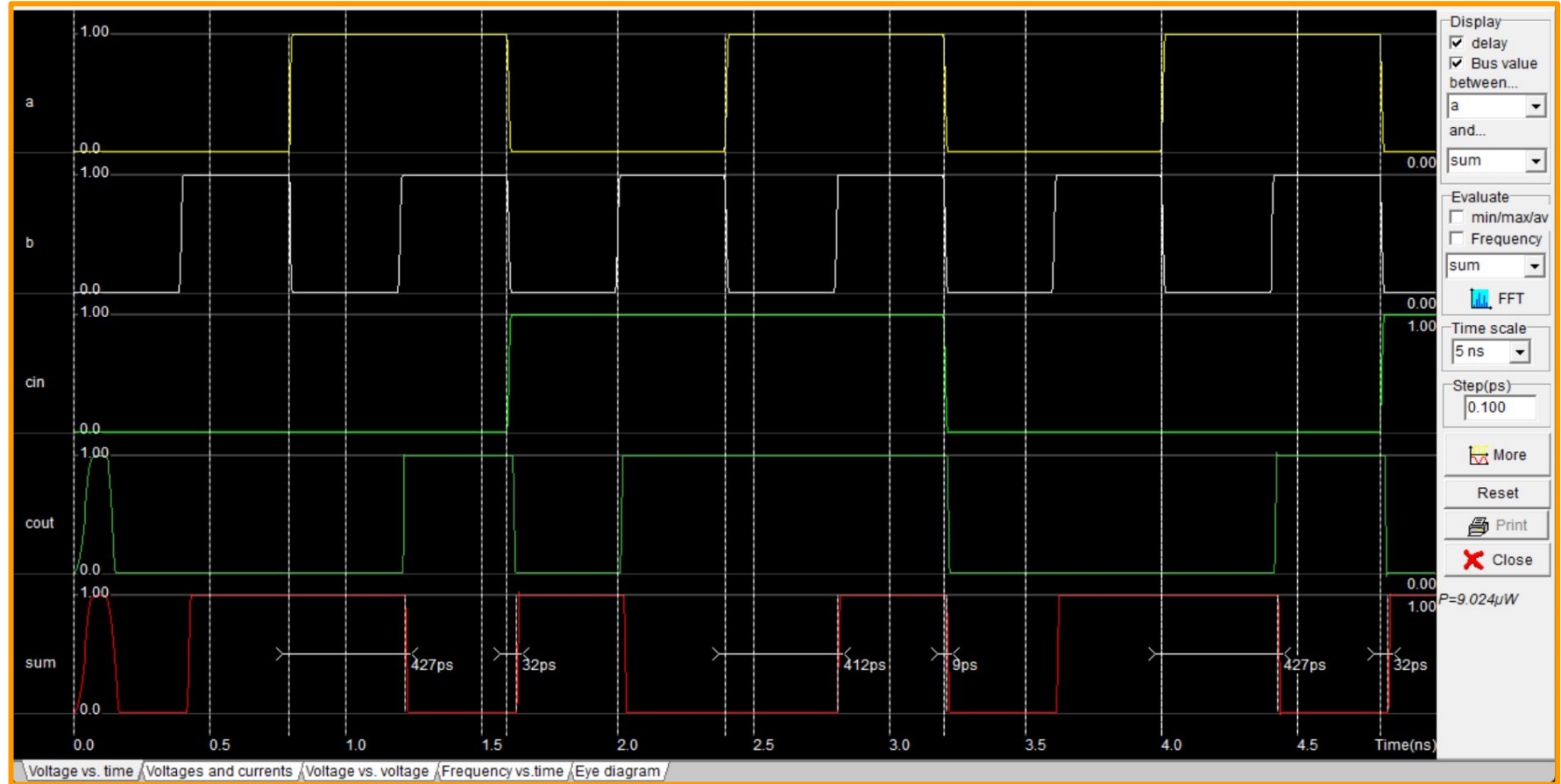


CMOS FA Layout

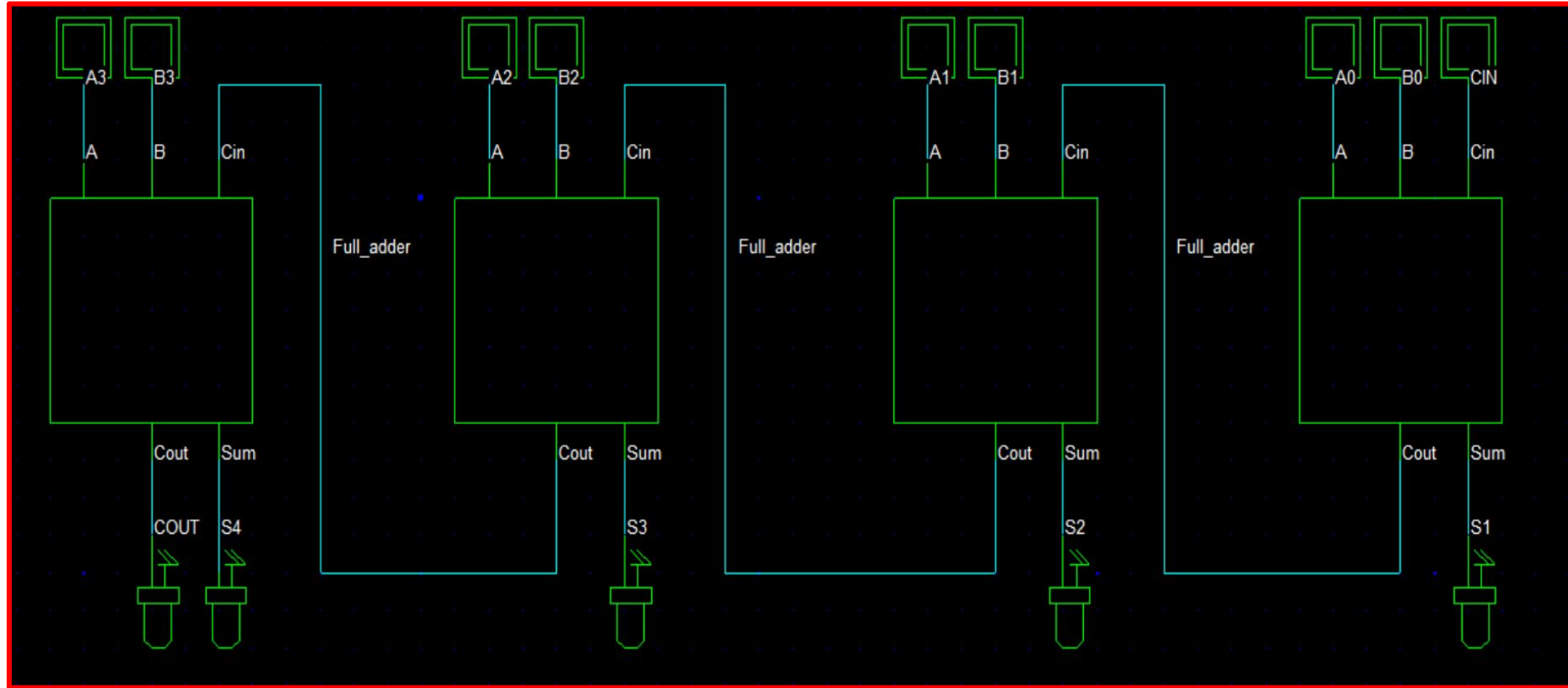
A: $0.790\text{ns} + 0.790\text{ns} = 1.680\text{ns}$ CIN: $1.590\text{ns} + 1.590\text{ns} = 3.180\text{ns}$ B: $0.390\text{ns} + 0.390\text{ns} = 0.780\text{ns}$



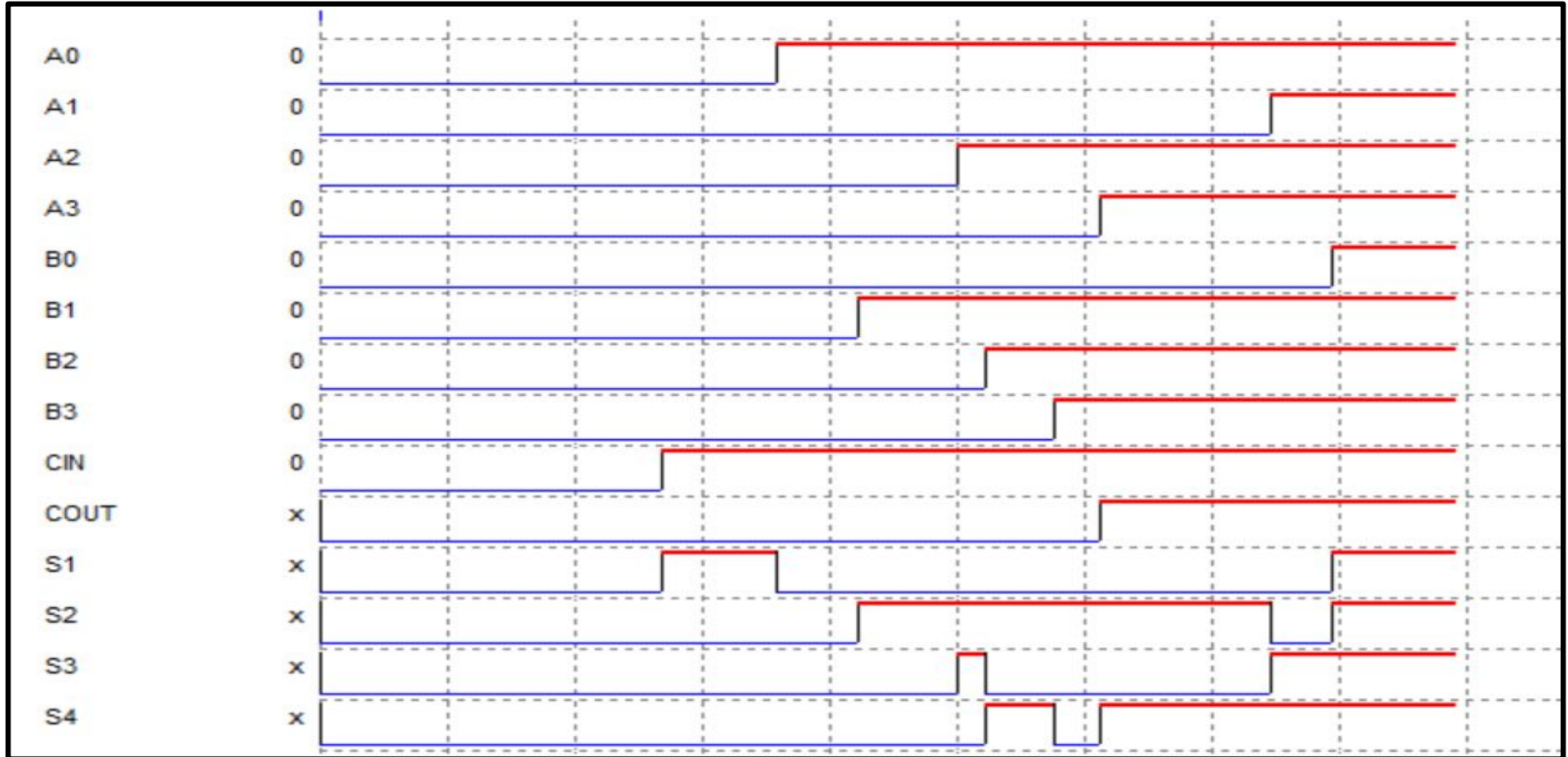
CMOS_FA_Layout Transient Analysis



CMOS FA 4Bit Schematic



CMOS FA 4Bit Schematic (Transient Analysis)



CMOS 4 Bit FA Layout

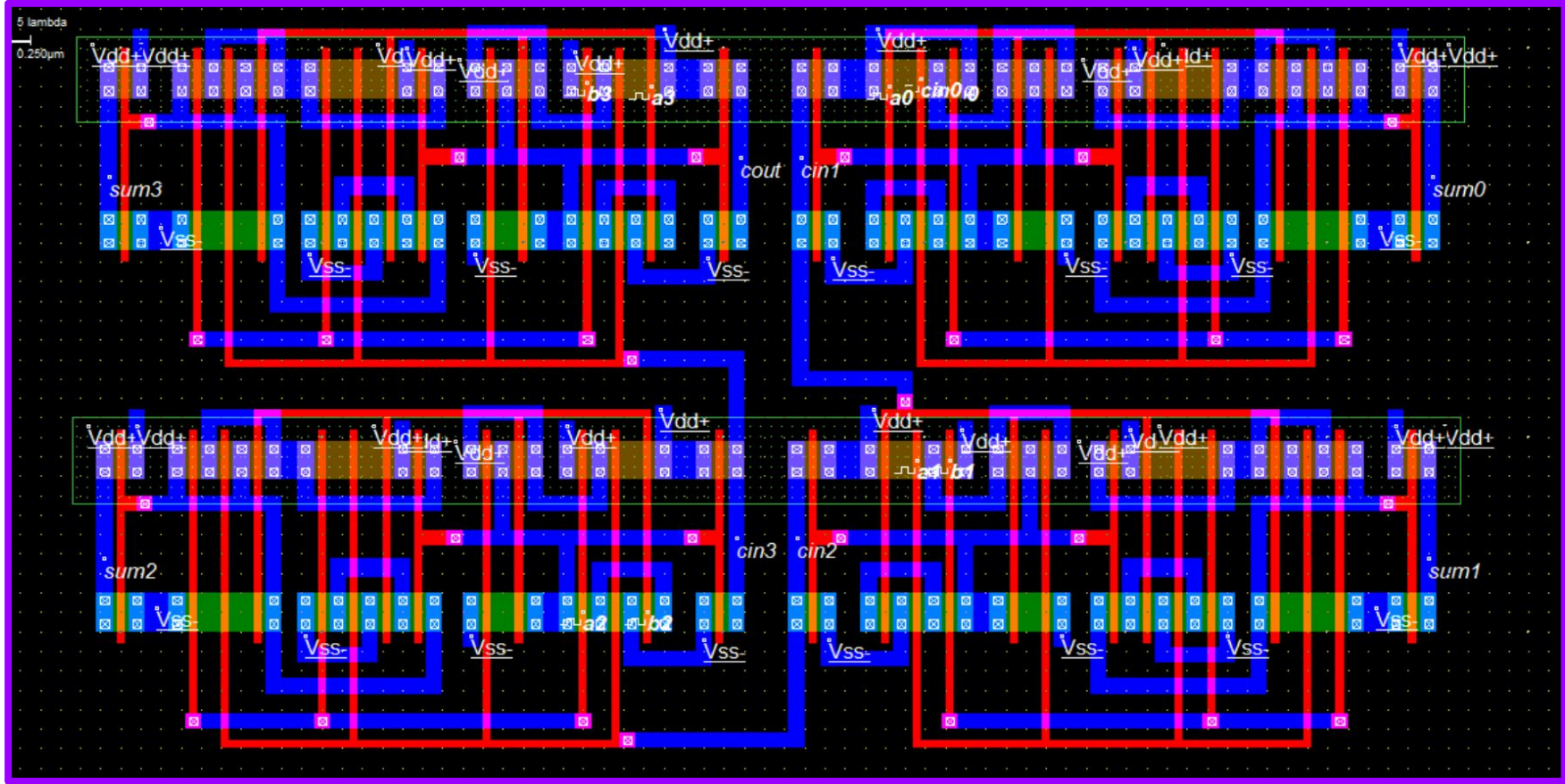
$$\text{CIN0: } 1.590\text{ns} + 1.590\text{ns} = 3.180\text{ns}$$

$$\text{A0: } 0.790\text{ns} + 0.790\text{ns} = 1.680\text{ns} \quad \text{B0: } 0.390\text{ns} + 0.390\text{ns} = 0.780\text{ns}$$

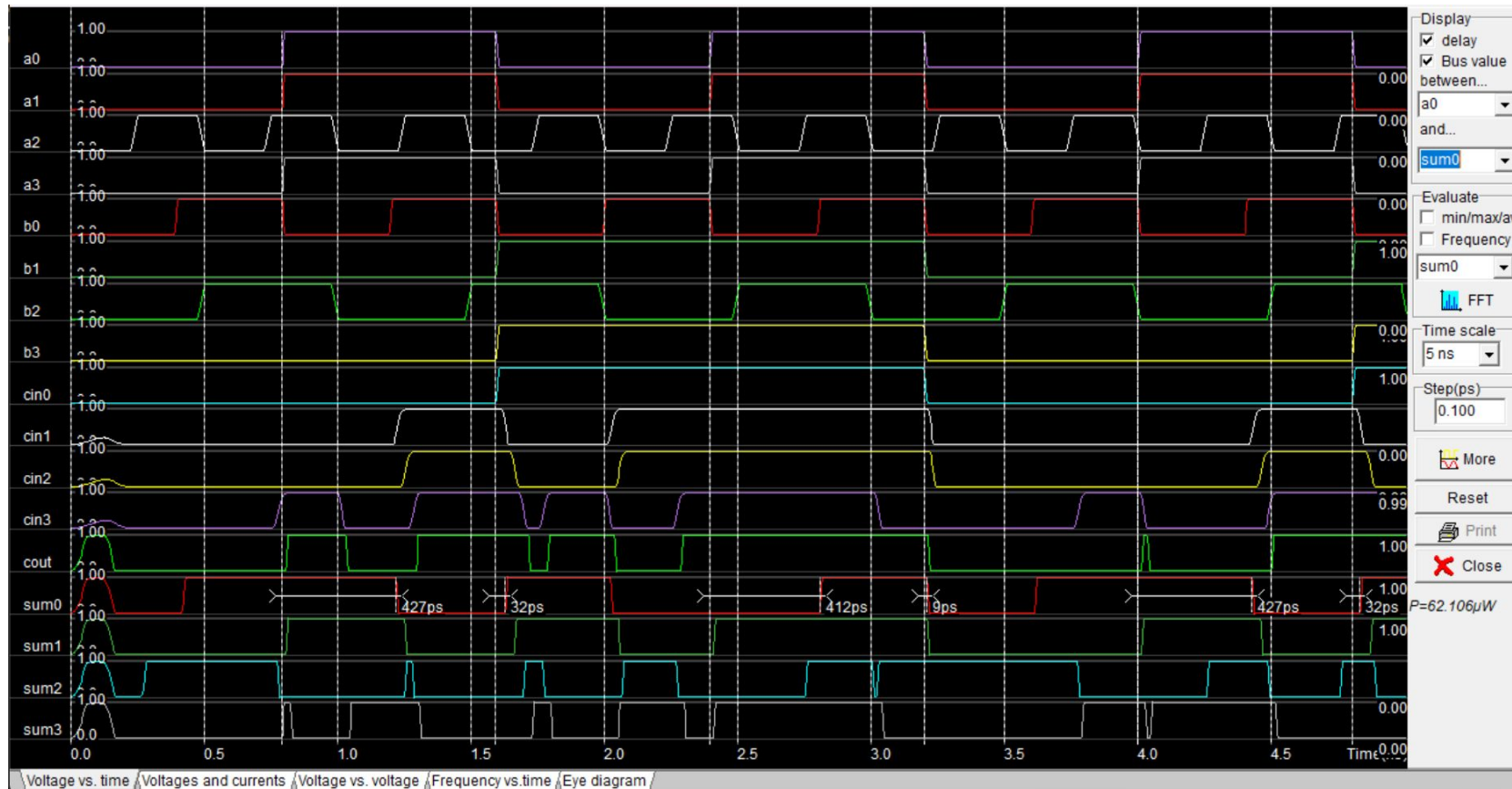
$$\text{A1: } 0.790\text{ns} + 0.790\text{ns} = 1.680\text{ns} \quad \text{B1: } 1.590\text{ns} + 1.590\text{ns} = 3.180\text{ns}$$

$$\text{A3: } 0.790\text{ns} + 0.790\text{ns} = 1.680\text{ns} \quad \text{B3: } 1.590\text{ns} + 1.590\text{ns} = 3.180\text{ns}$$

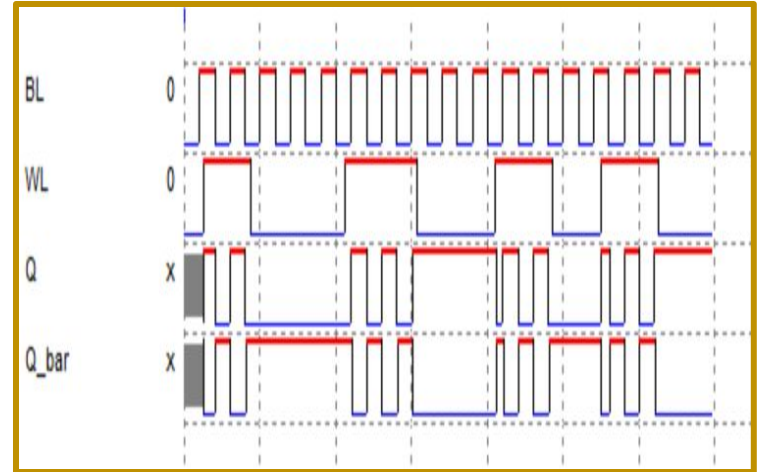
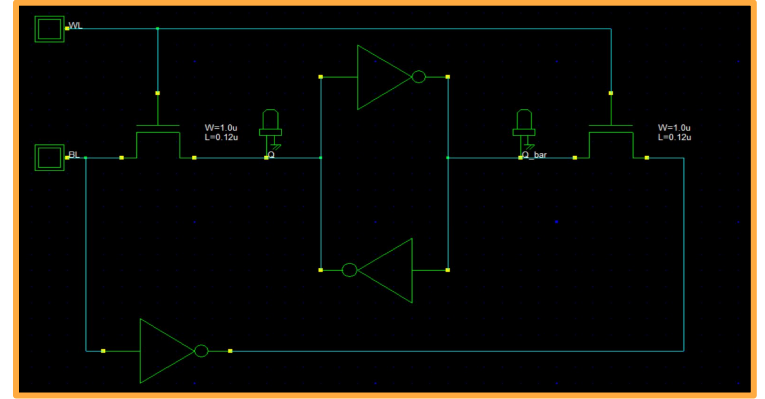
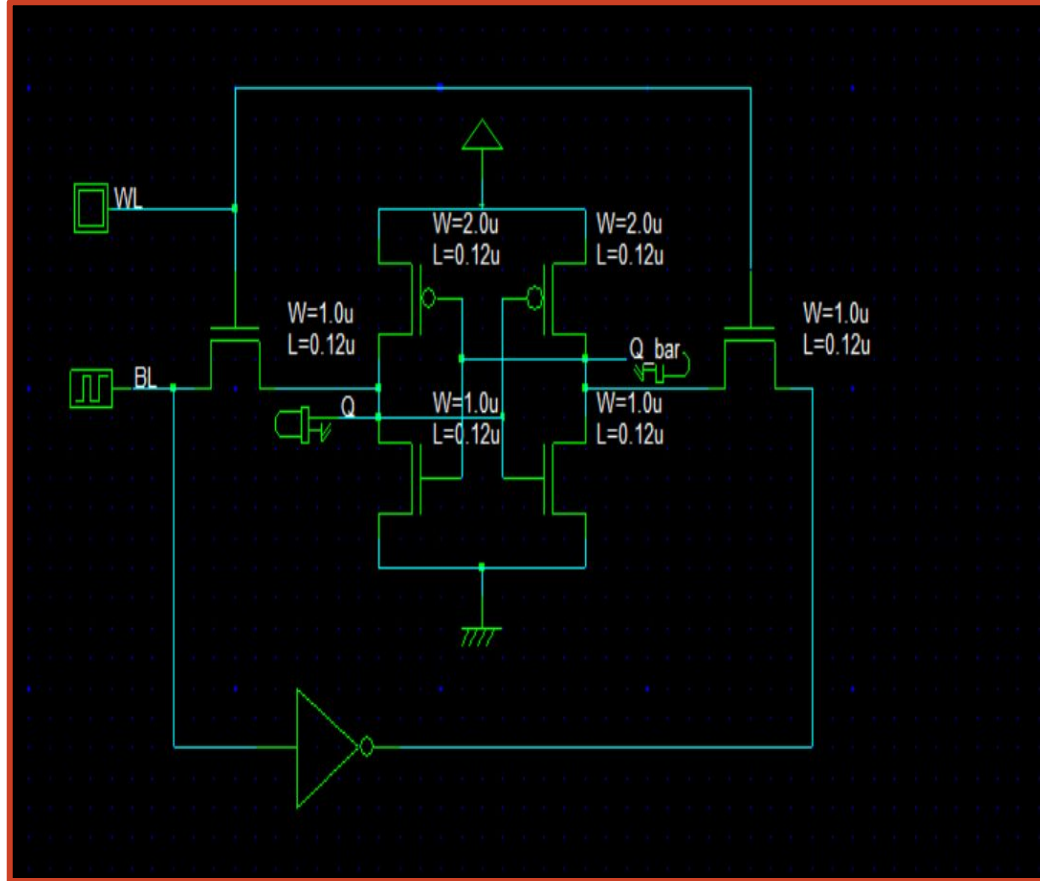
$$\text{A2: } 0.225\text{ns} + 0.225\text{ns} = 0.450\text{ns} \quad \text{B2: } 0.475\text{ns} + 0.475\text{ns} = 0.950\text{ns}$$



CMOS 4 Bit FA Layout Transient Analysis

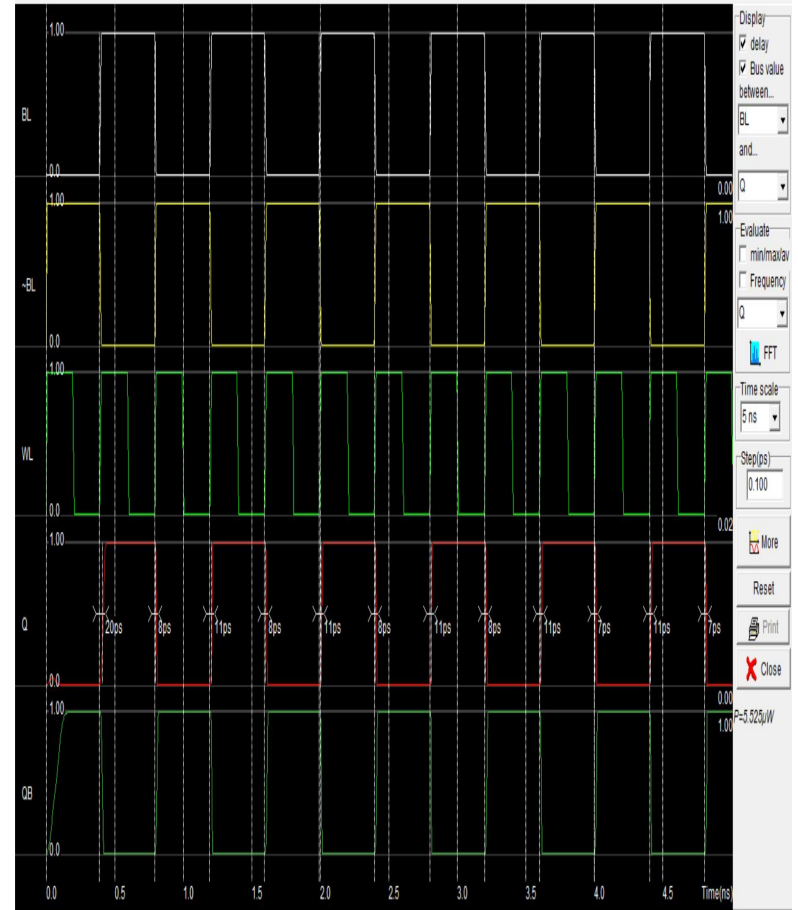
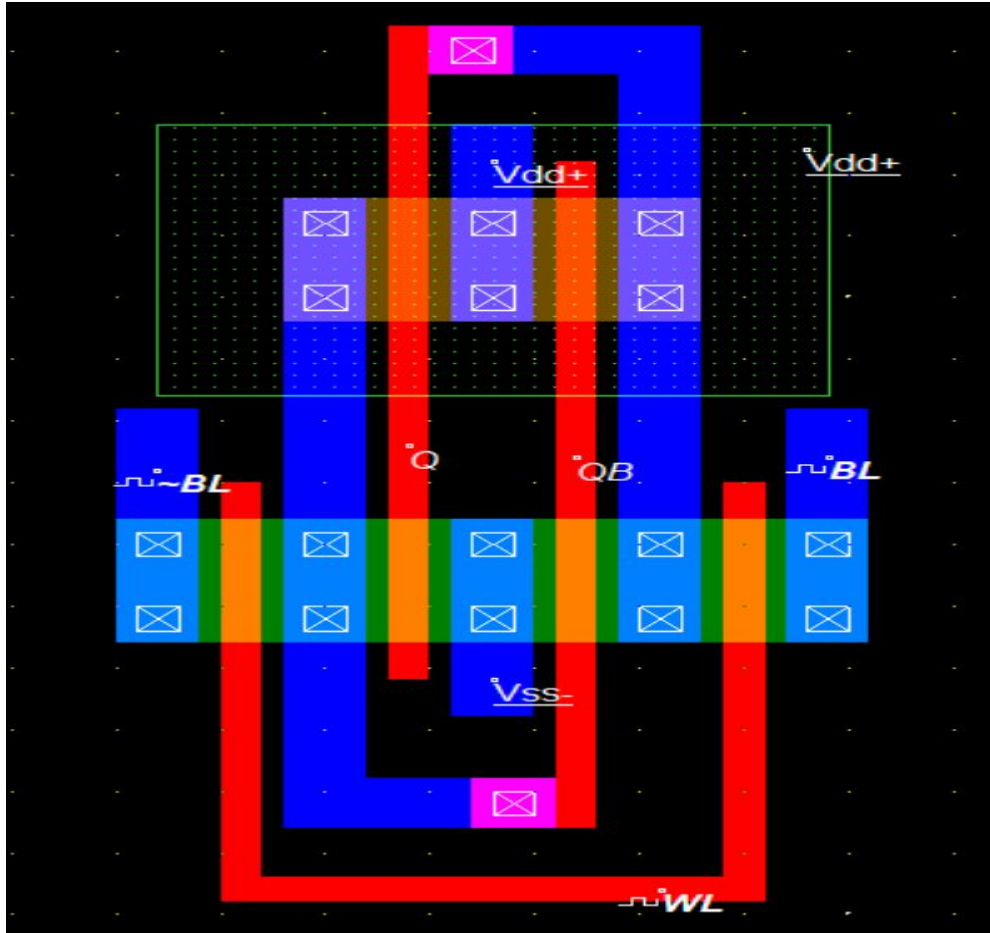


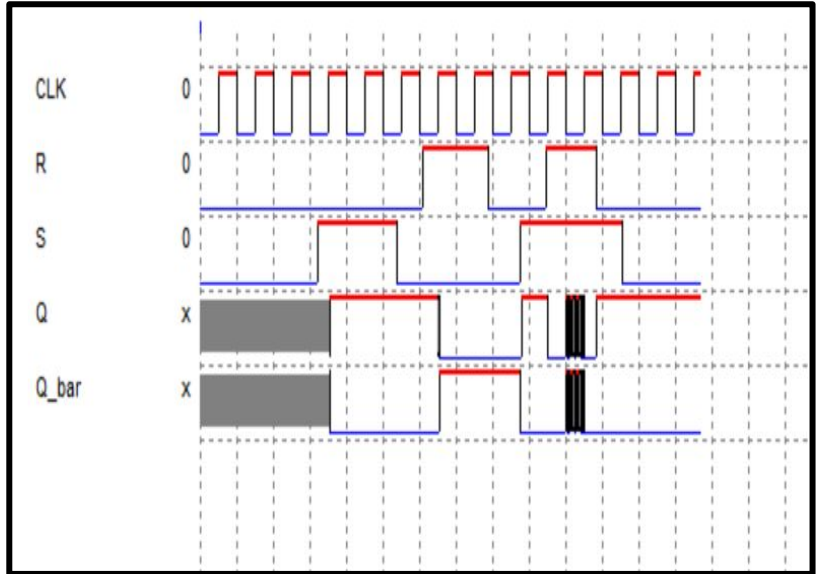
CMOS 6T SRAM Schematic



CMOS 6T1RAM Layout

BL : $0.390\text{ns} + 0.390\text{ns} = 0.780\text{ns}$ WL : $0.190\text{ns} + 0.190\text{ns} = 0.380\text{ns}$

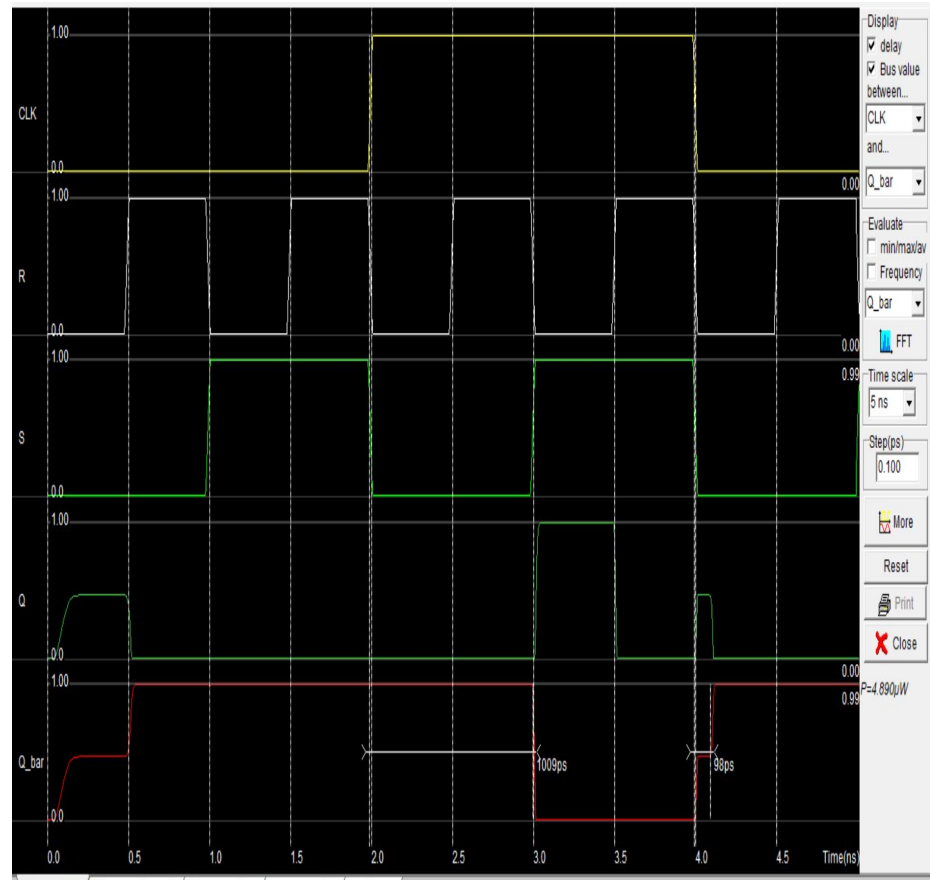
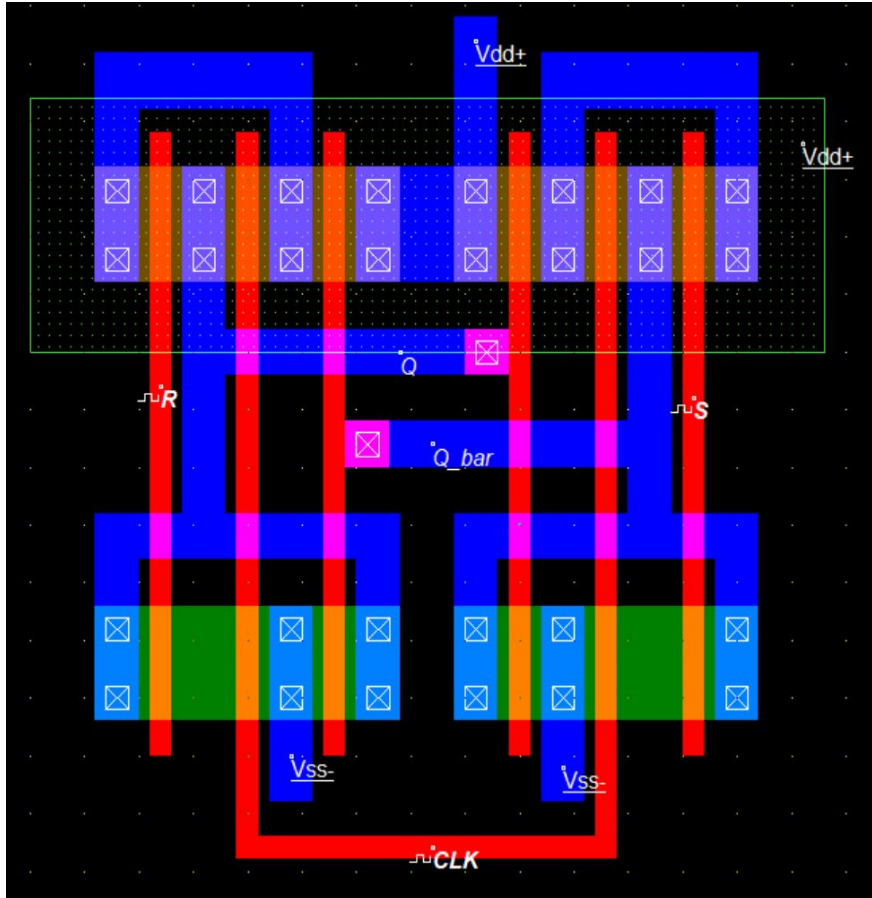




CMOS SRFF Layout

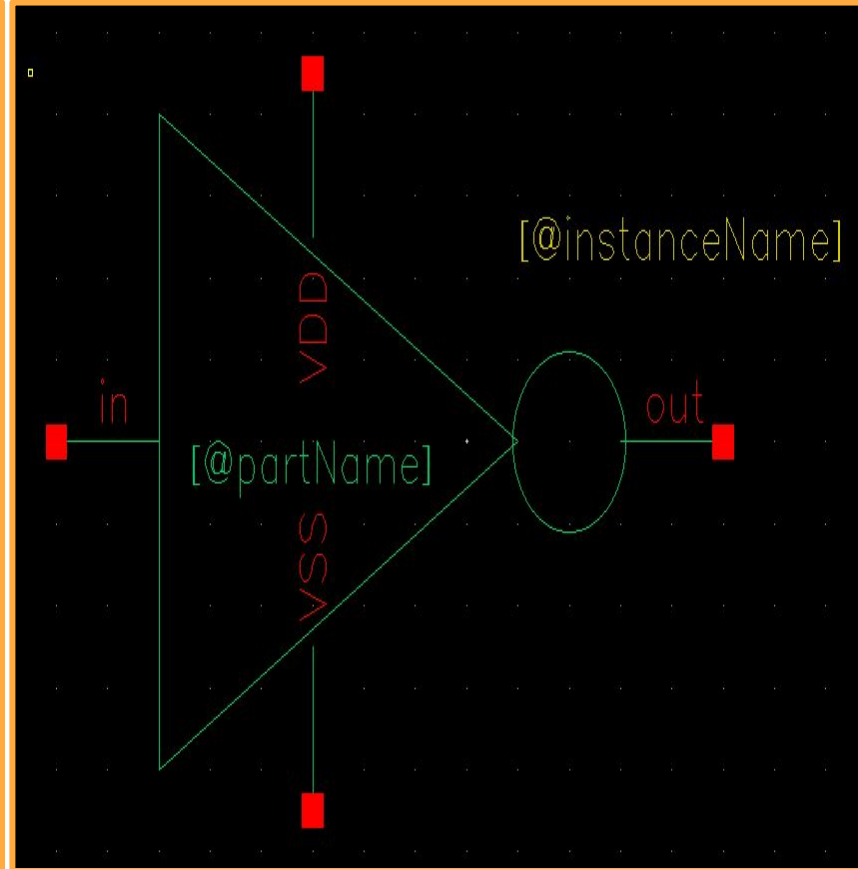
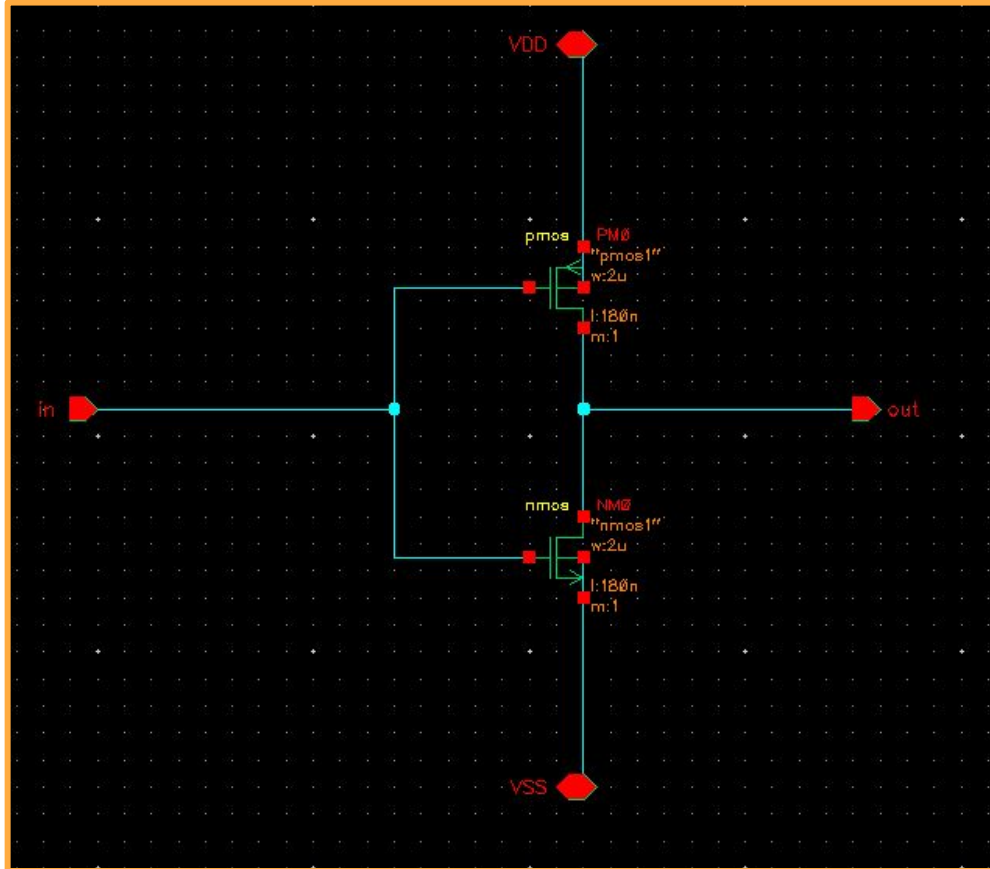
R: $0.475\text{ns} + 0.475\text{ns} = 0.950\text{ns}$ CLK: $1.975\text{ns} + 1.975\text{ns} = 3.950\text{ns}$

S: $0.975\text{ns} + 0.975\text{ns} = 1.950\text{ns}$

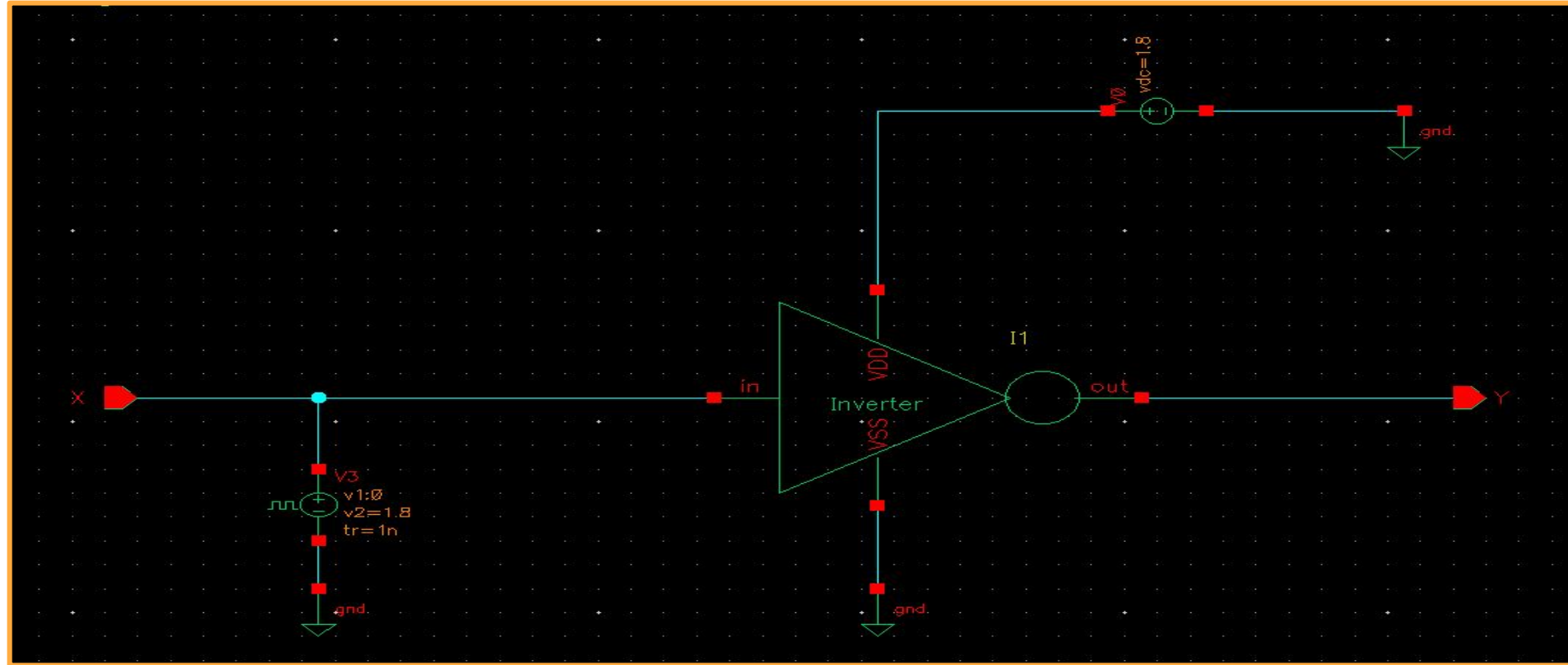


CADENCE LABS

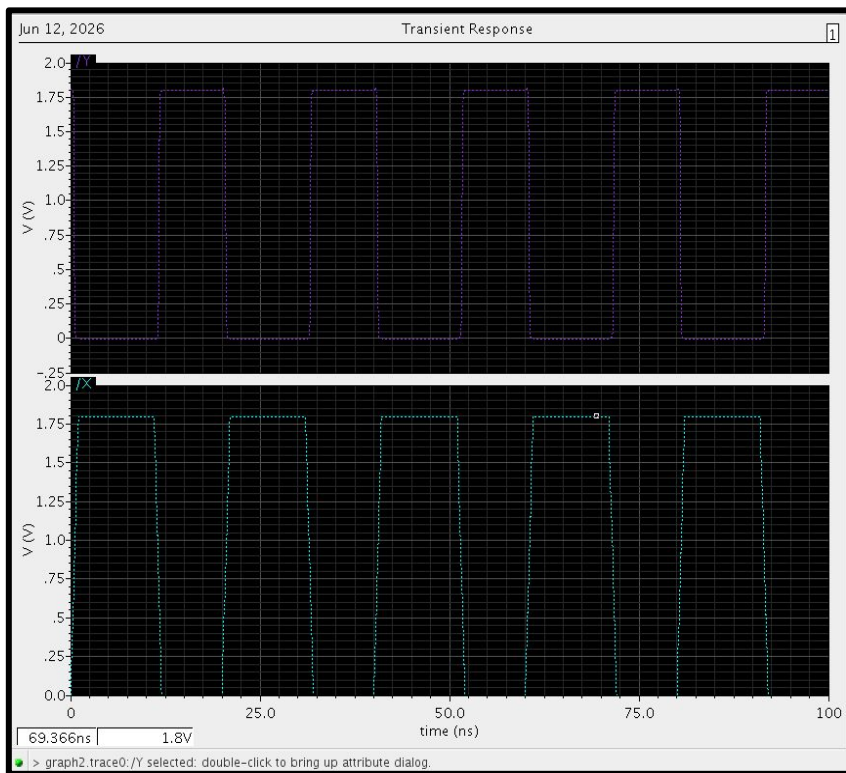
Inverter Schematic and Symbol



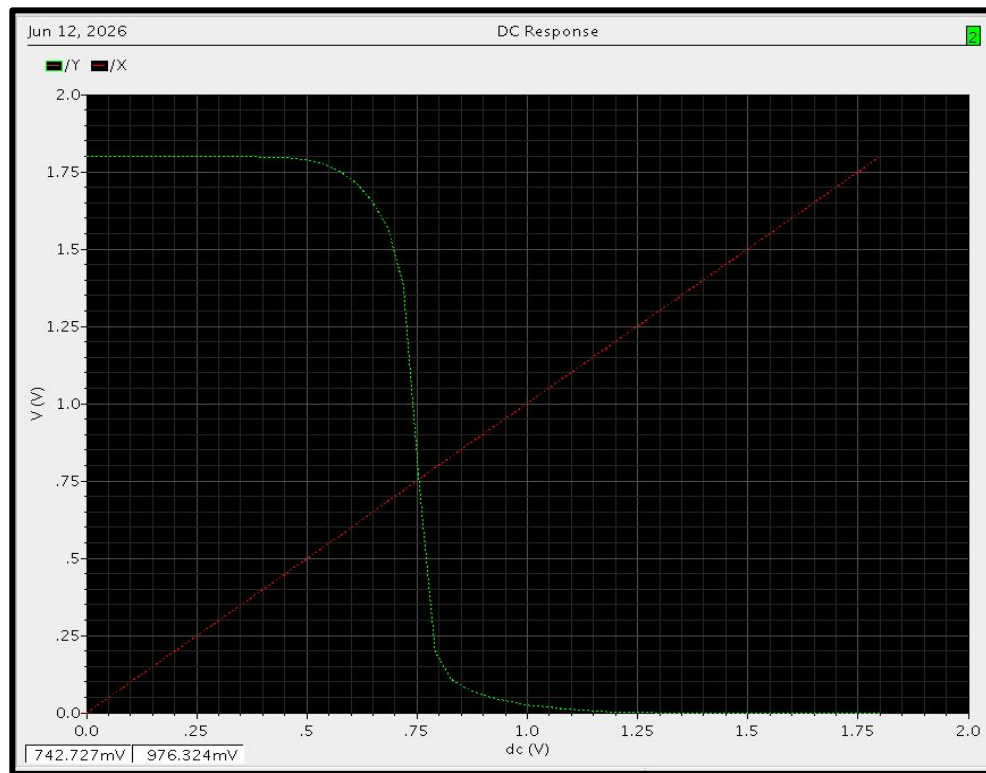
Inverter Test Bench



Inverter Transient/DC Analysis

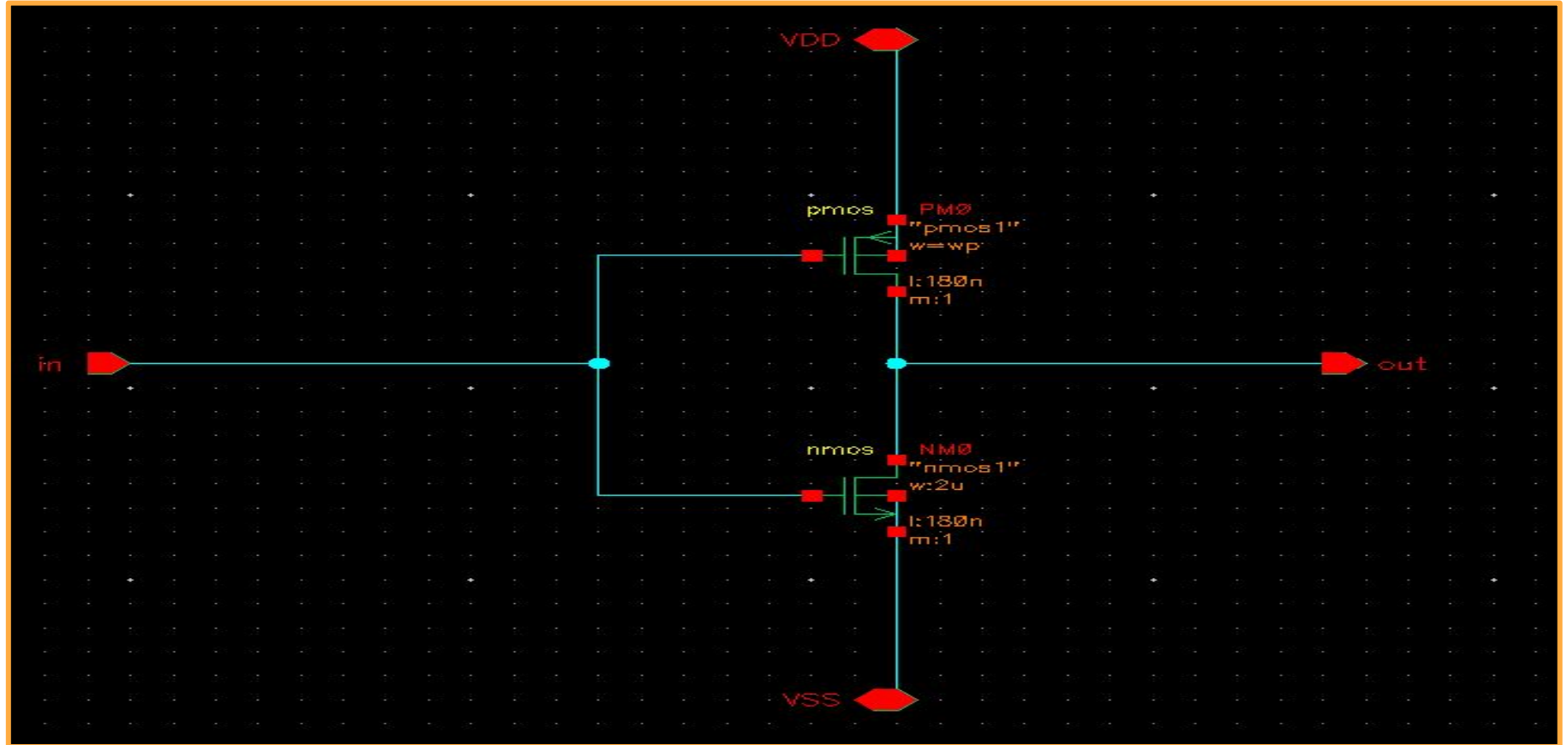


Transient Analysis

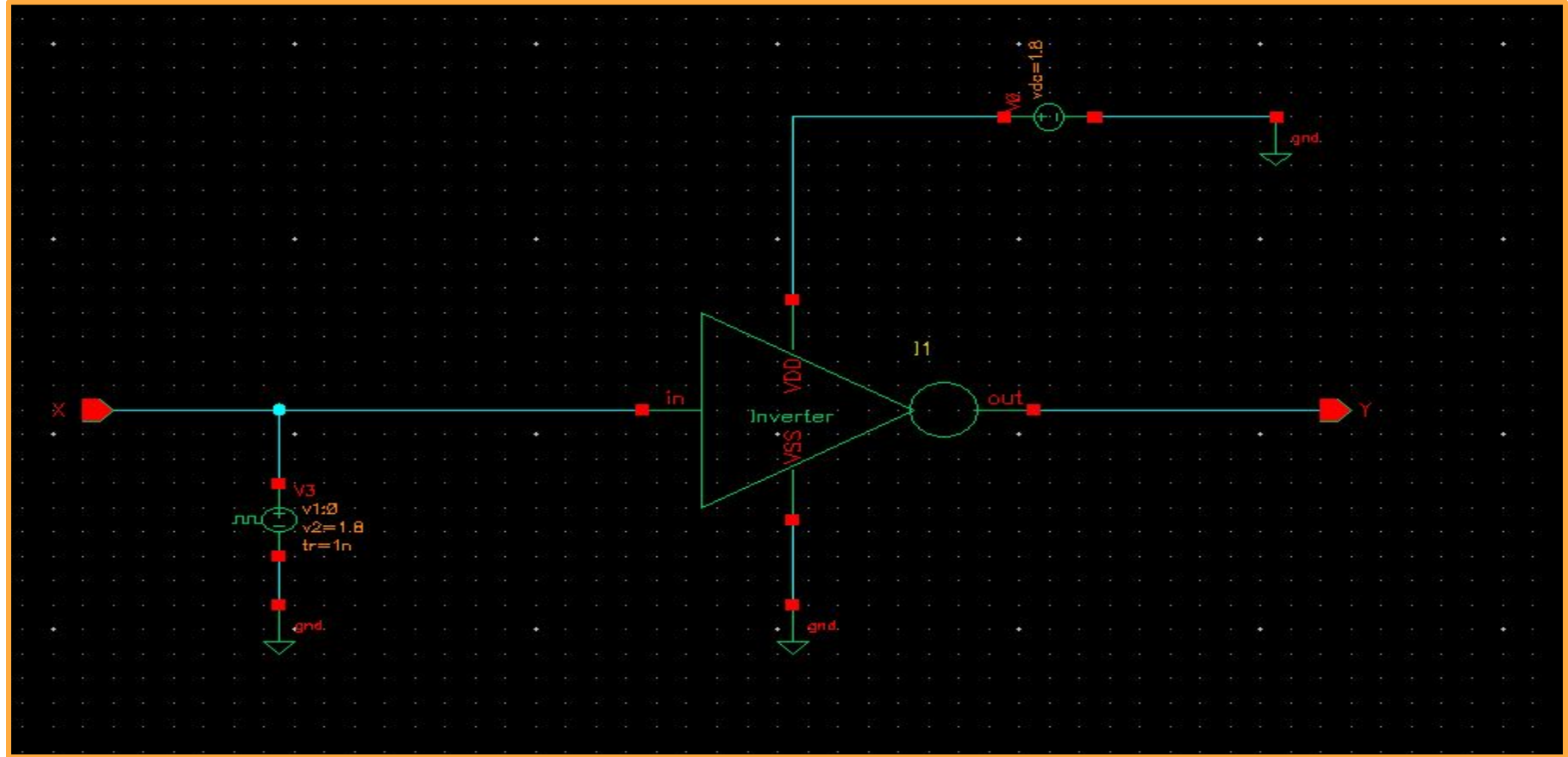


DC Analysis

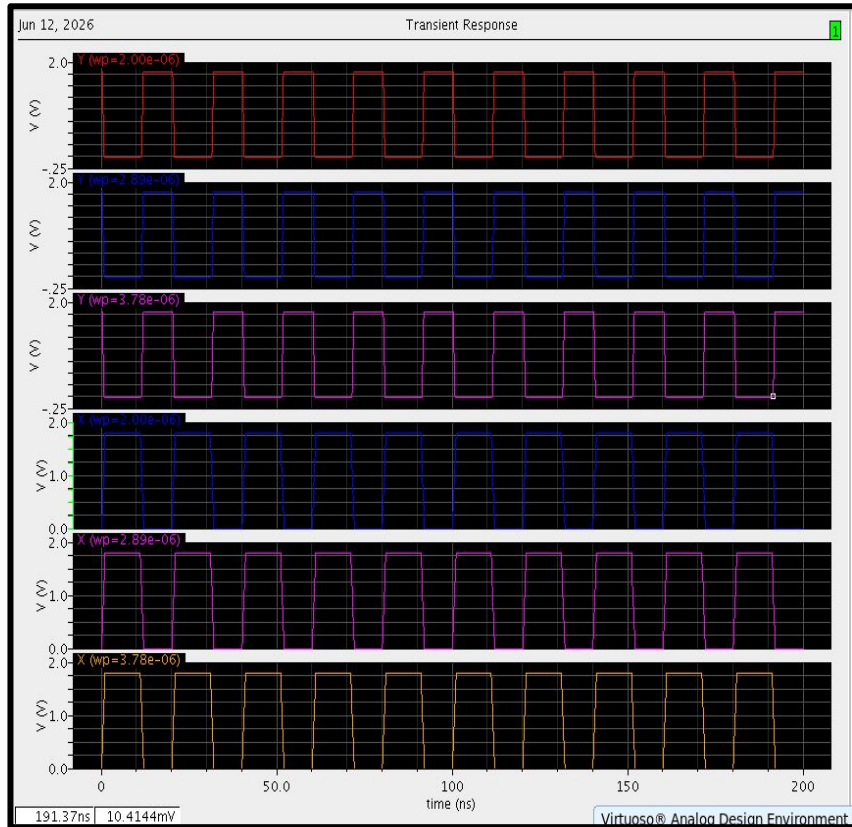
Inverter (Parameterized) Schematic



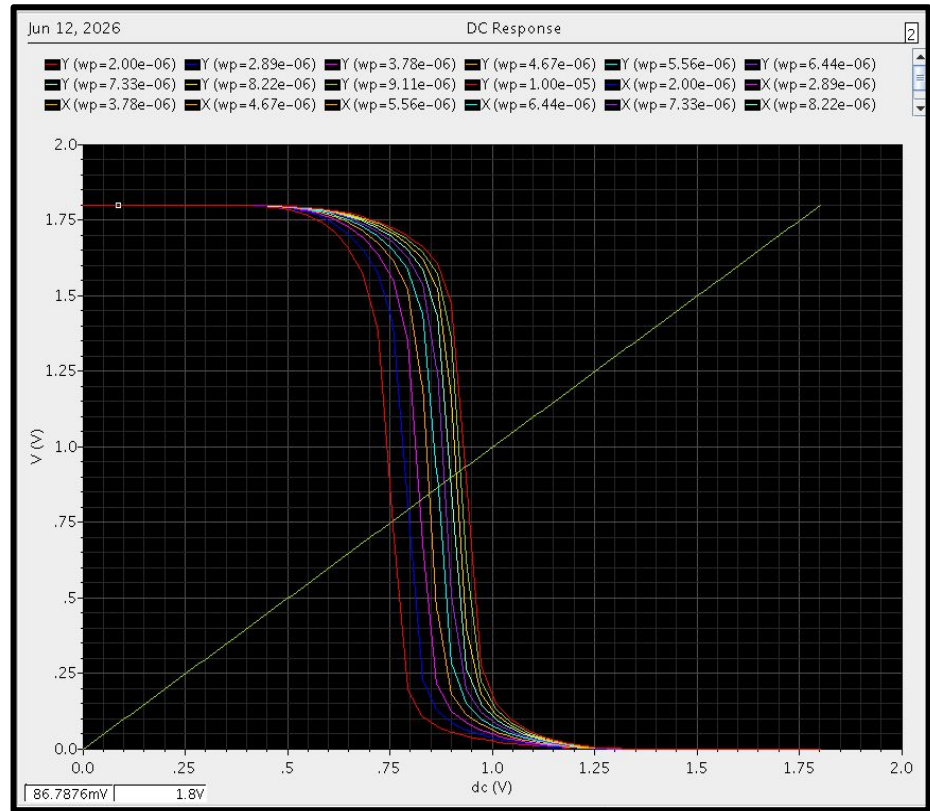
Inverter (Para) Test Bench



Inverter (Para) Transient/DC Analysis

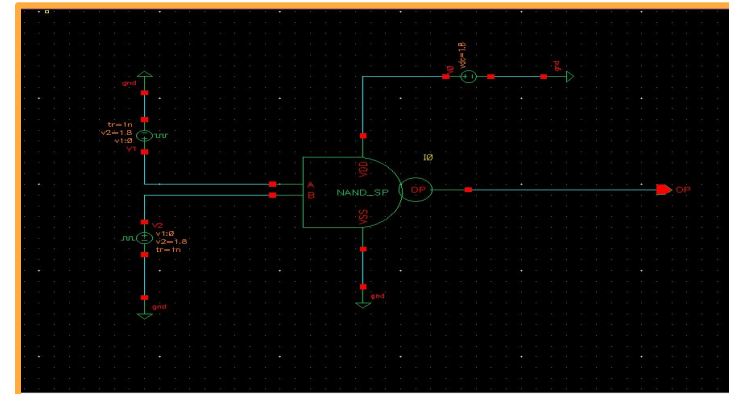
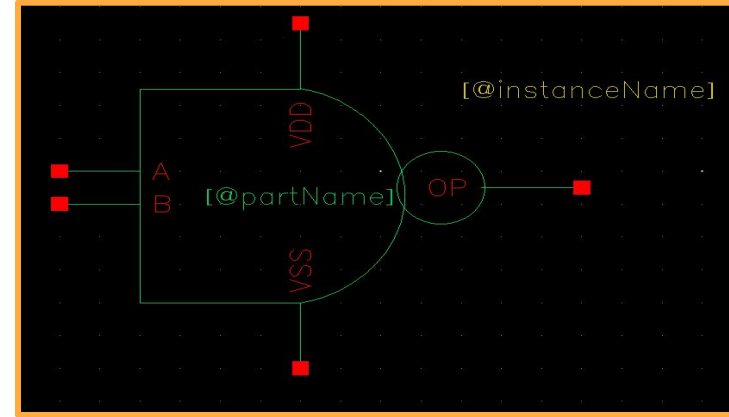
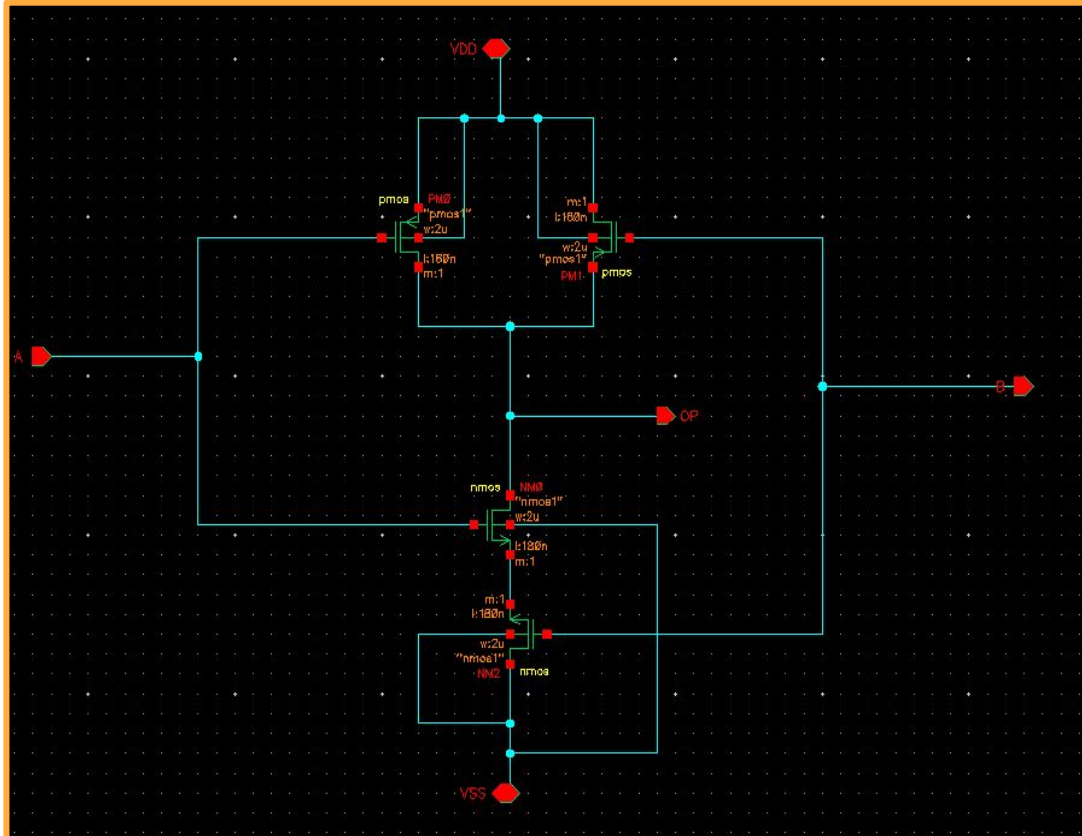


Transient Analysis

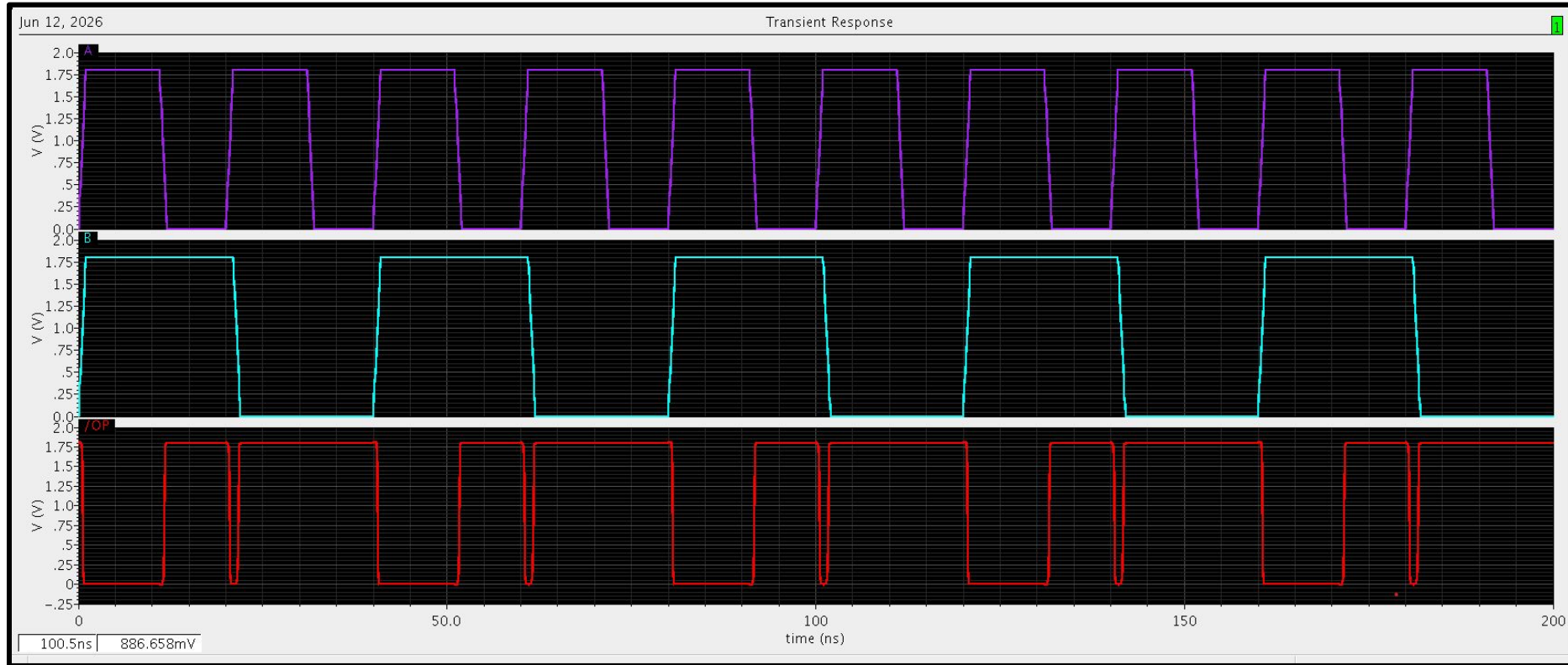


DC Analysis

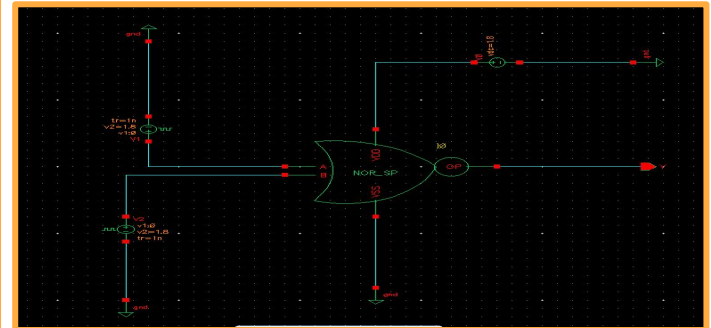
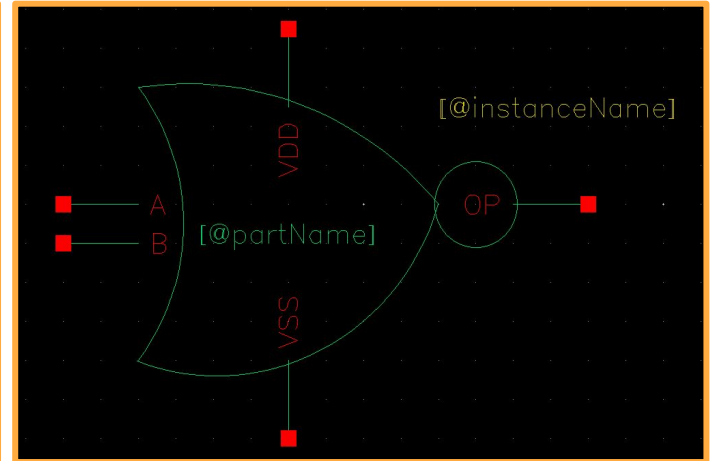
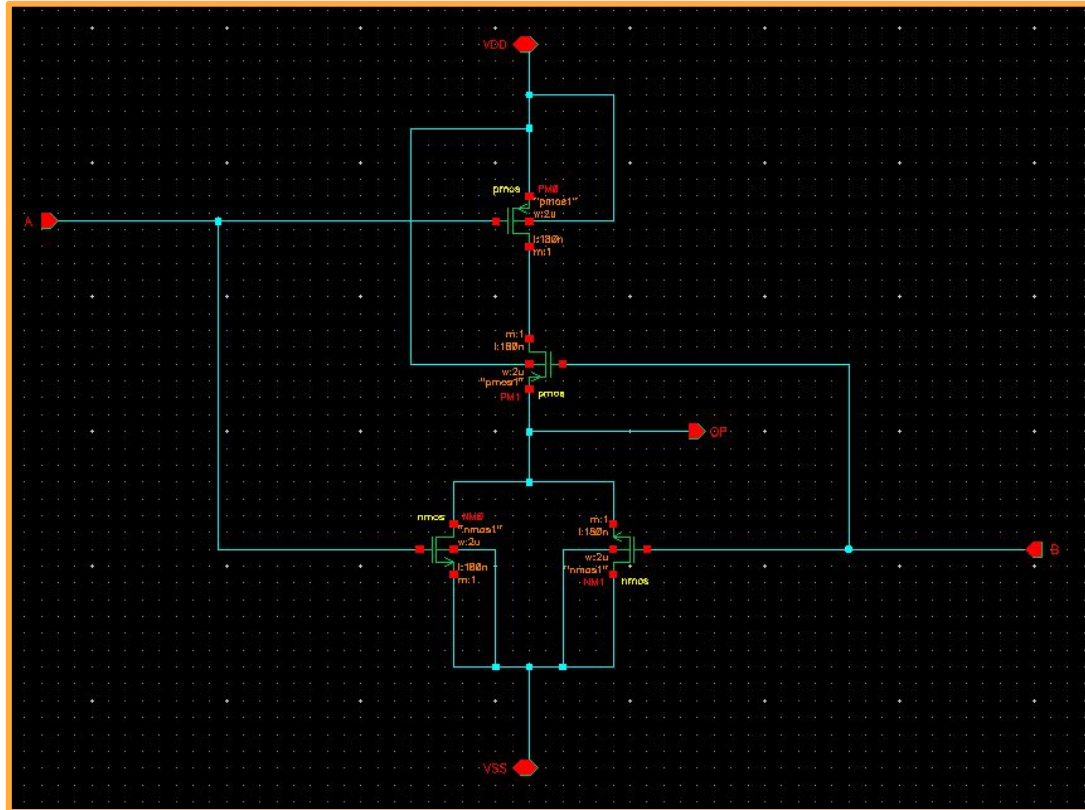
NAND Schematic, Symbol and TestBench



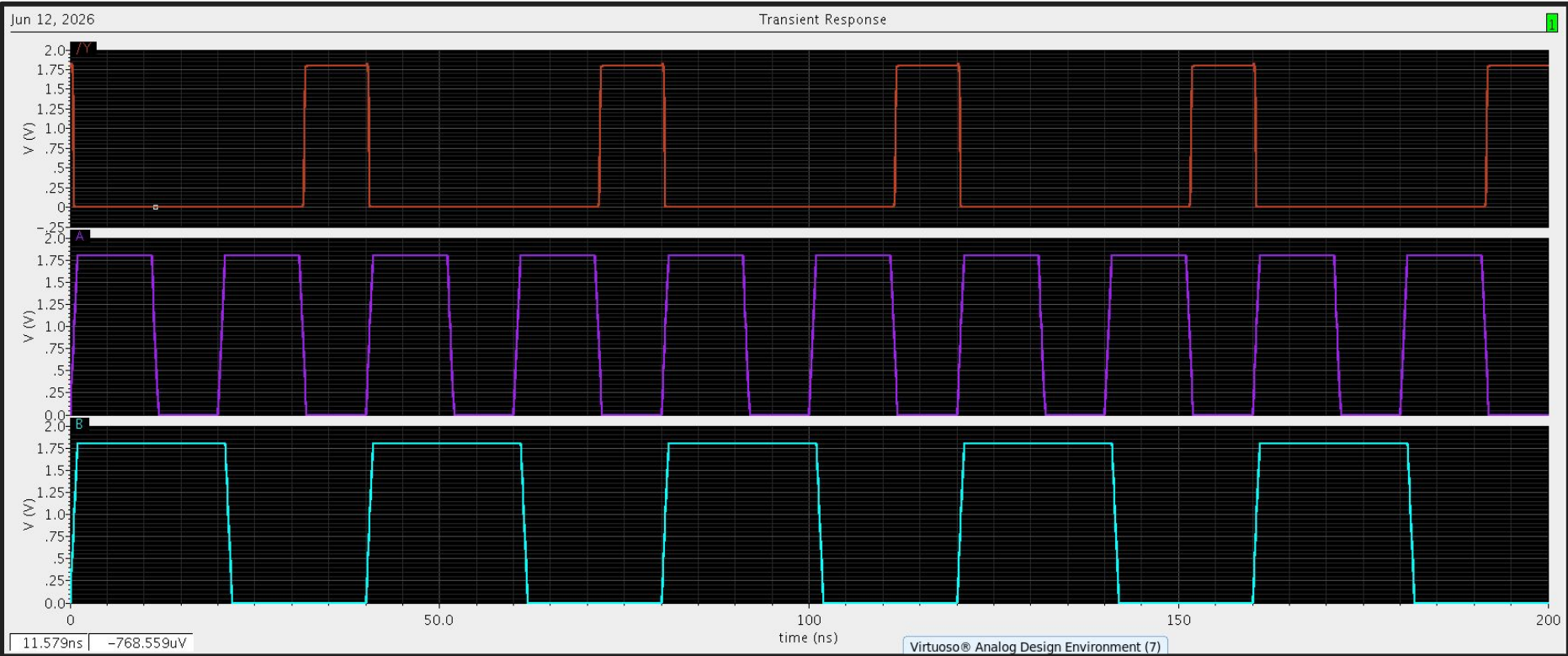
NAND Transient Analysis



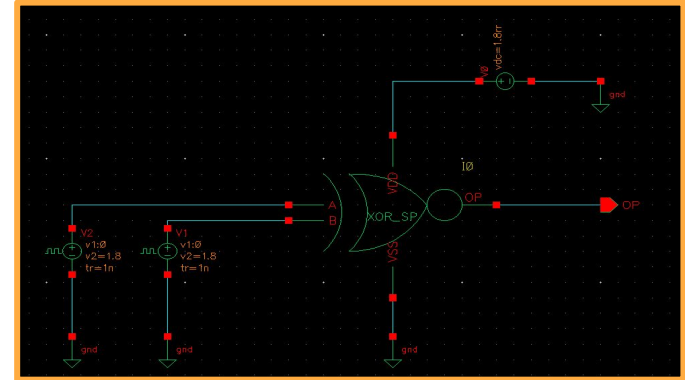
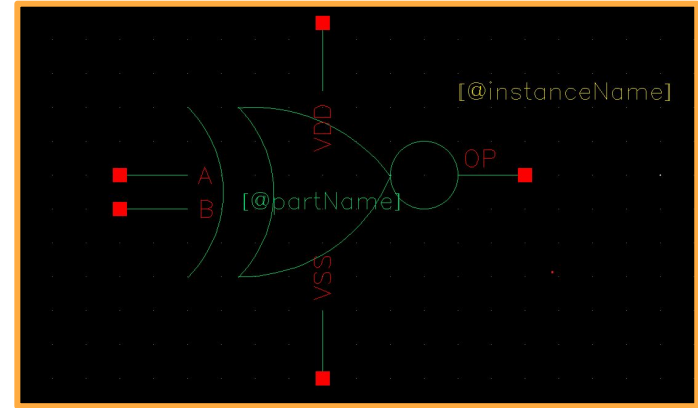
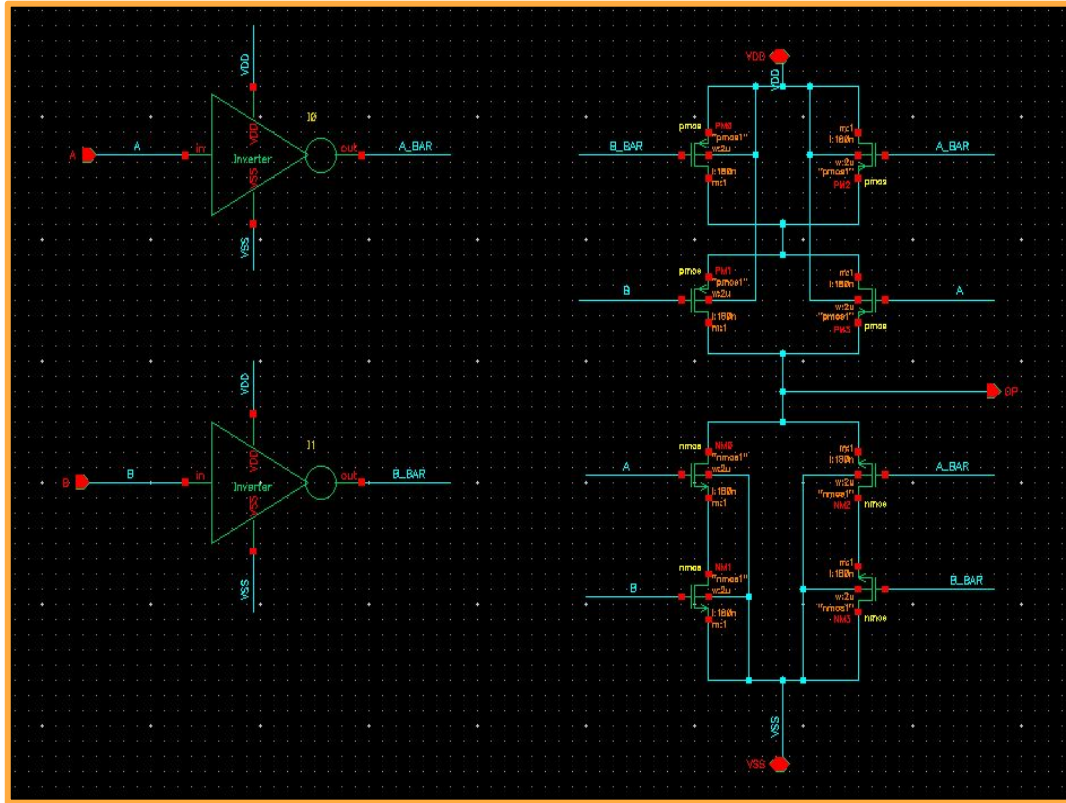
NOR Schematic, Symbol and TestBench



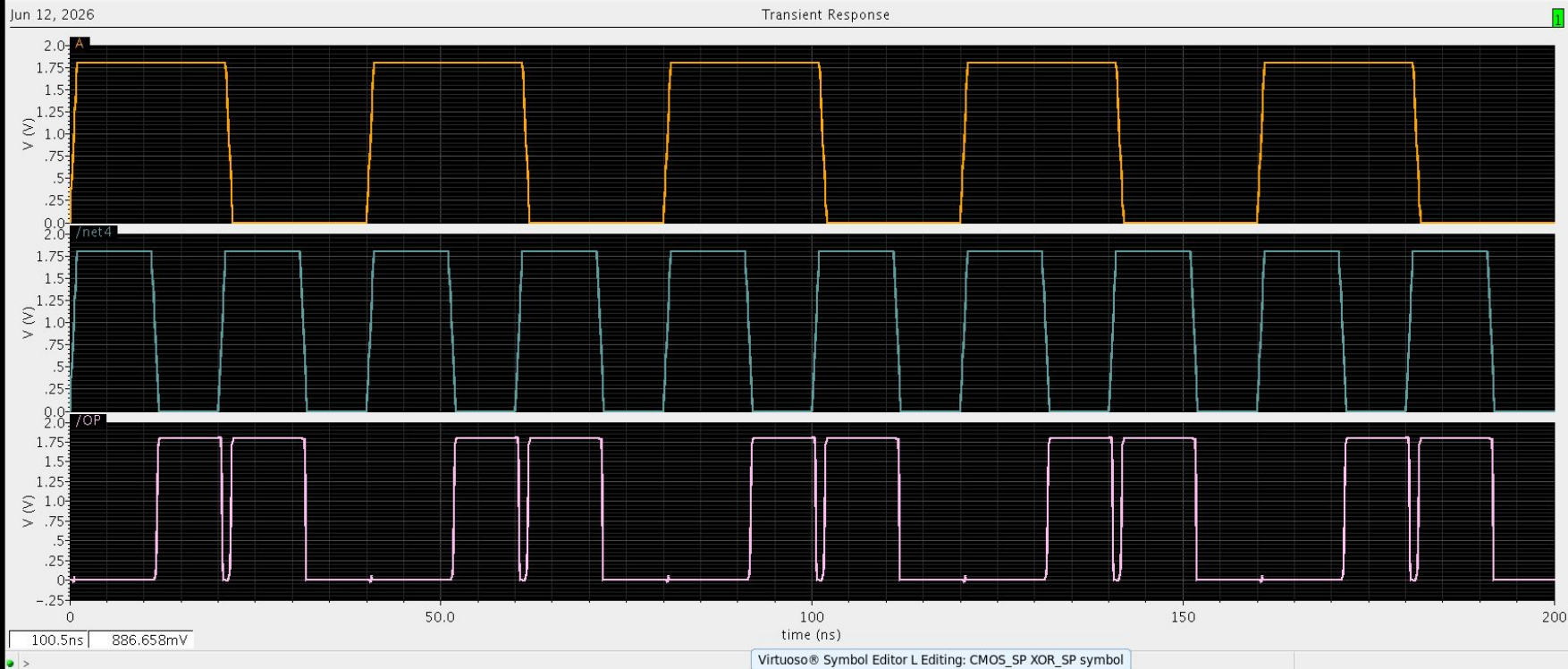
NOR Transient Analysis



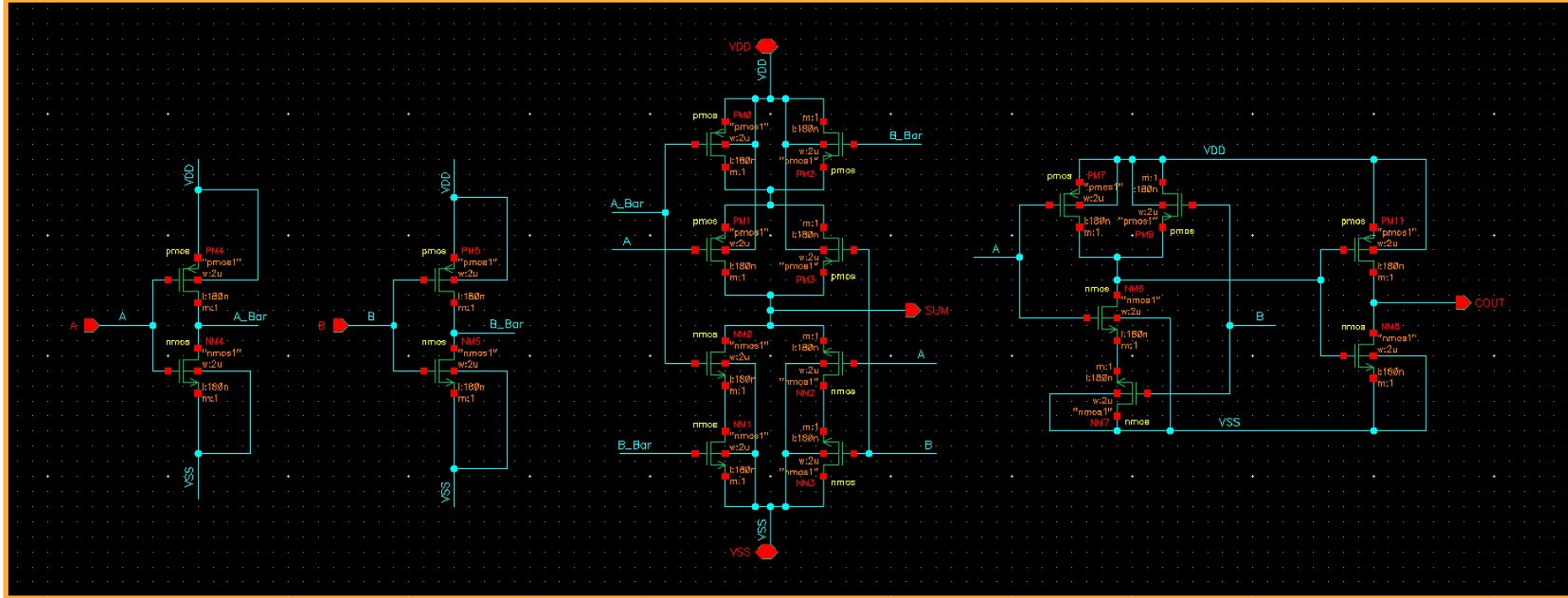
XOR Schematic, Symbol and TestBench



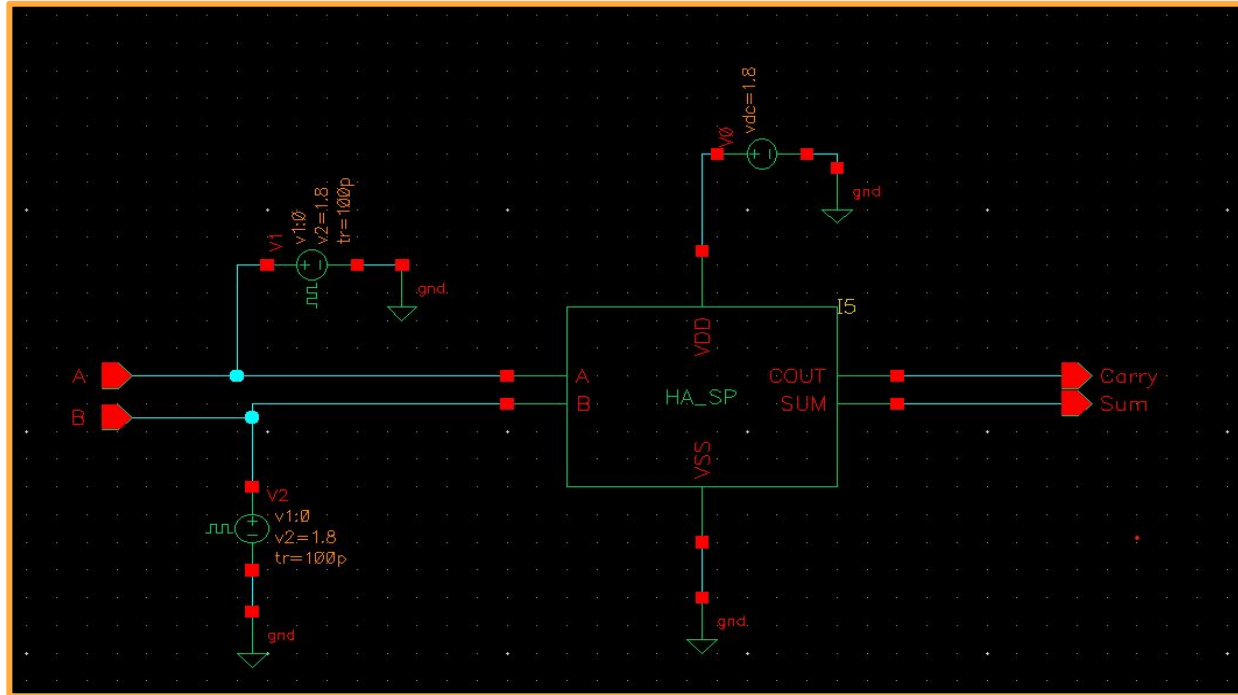
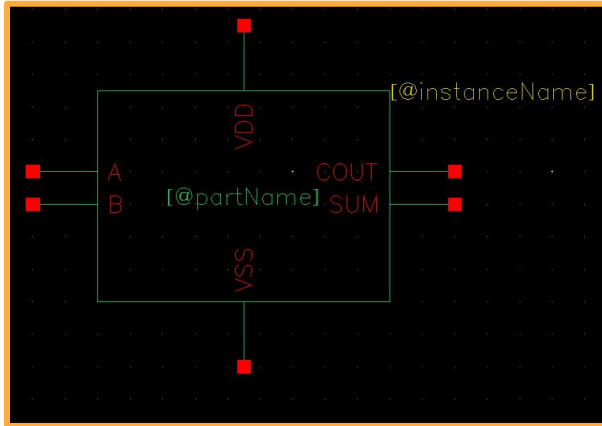
XOR Transient Analysis



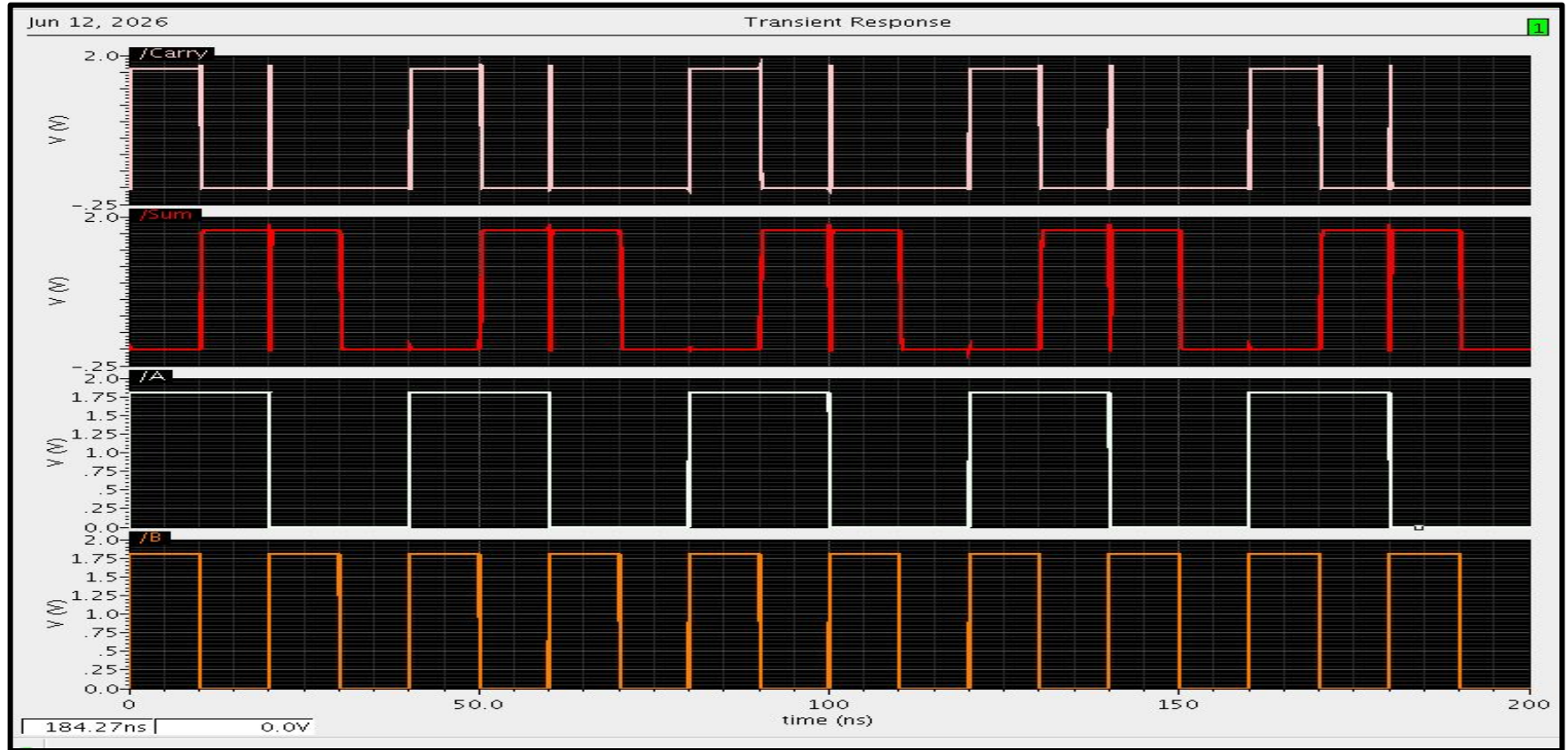
Half Adder Schematic



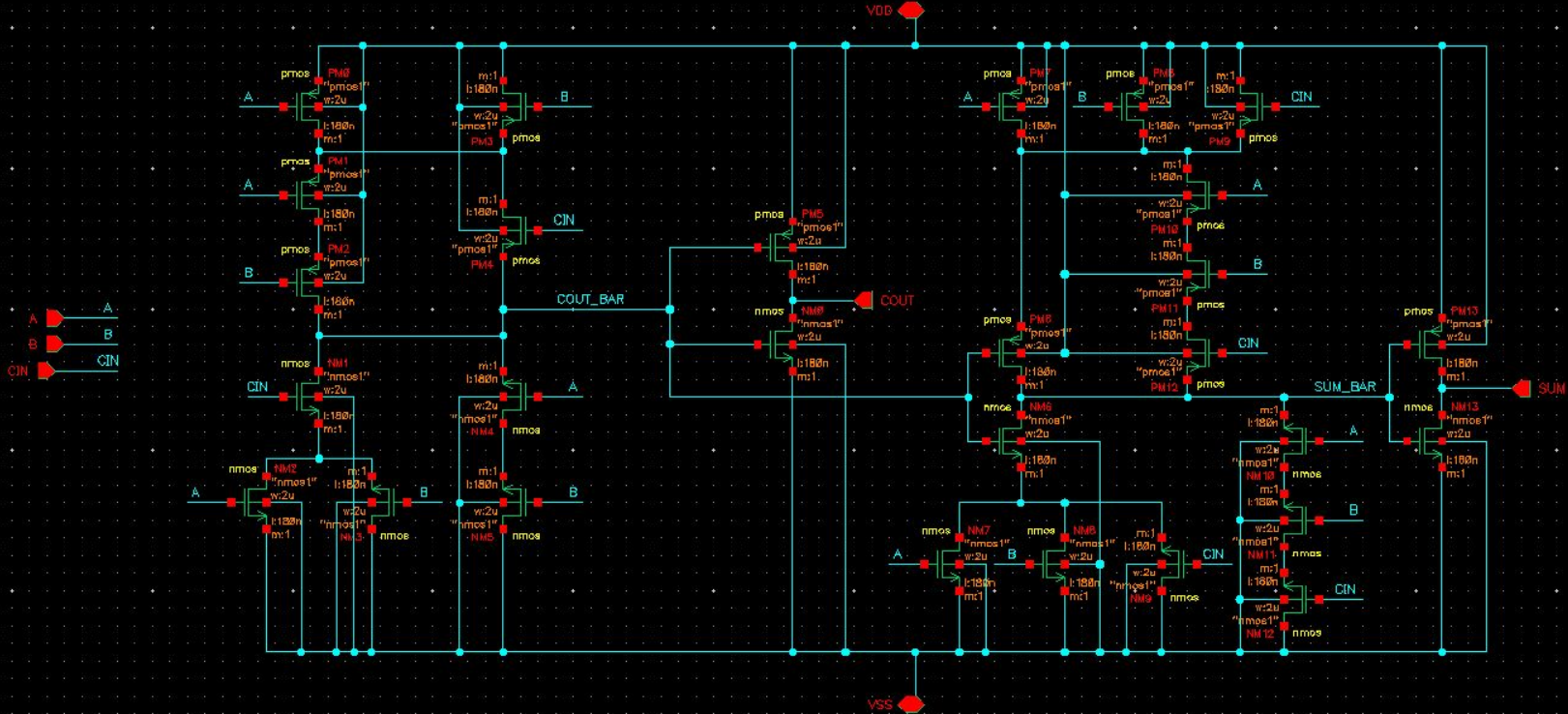
Half Adder Symbol and TestBench



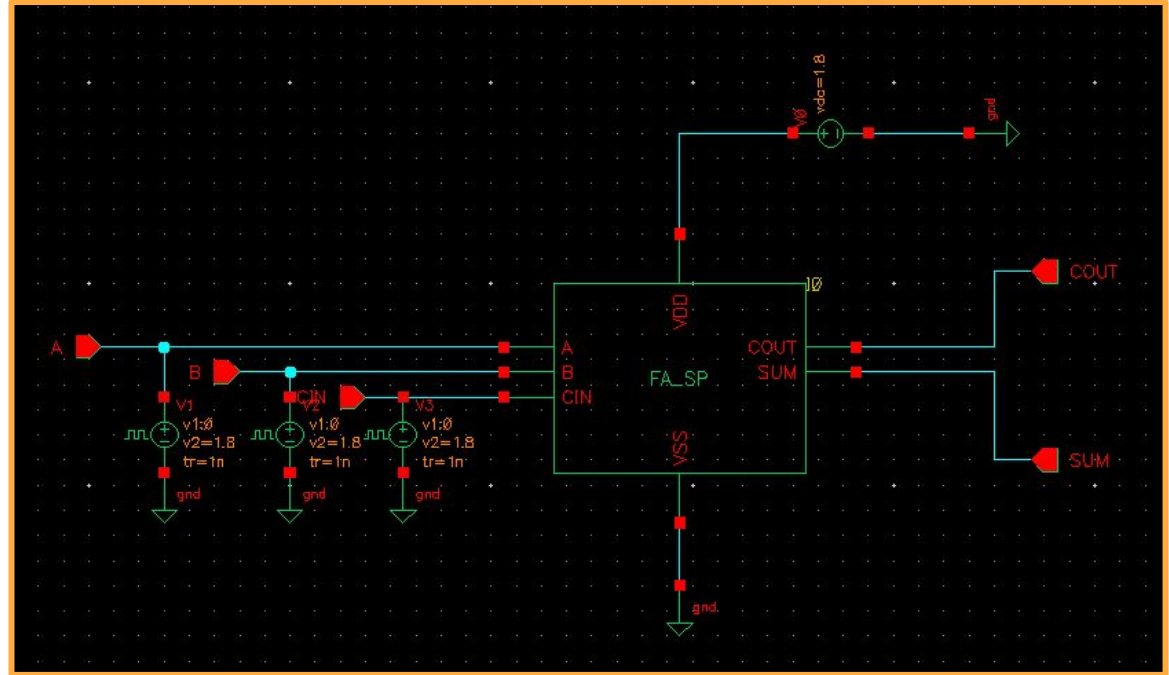
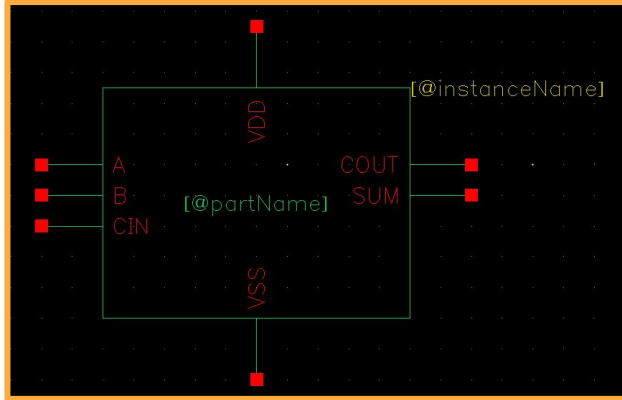
Half Adder Transient Analysis



Full Adder Schematic



Full Adder Symbol and TestBench



Full Adder Transient Analysis

Jun 13, 2026

Transient Response

